ALMA MATER STUDIORUM
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Theoretical Physics

Quantum Field Theory II

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Contents

Introduction

In these notes, we will explore the fascinating world of Quantum Field Theory (QFT). QFT is a fundamental framework in physics that combines classical field theory, special relativity, and quantum mechanics. It provides a powerful language for describing the behavior of particles and fields at the most fundamental level. We assume are going to assume that the reader has a basic understanding of quantum mechanics and special relativity, as well as some familiarity with classical field theory and quantum field theory of free fields (Which will aniway be reviewed in the first chapter).

There exist different motivations for studying QFT: it provides a framework for describing the interactions between particles and fields, and it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. Additionally:

- QFT is the language of the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity); it is used for the quantization of systems with an **infinite number of degrees of freedom**, such as fields; it is essential for understanding the behavior of particles at high energies and small distances, where quantum effects become significant; and it provides a framework for describing the behavior of particles in curved spacetime, which is important for understanding phenomena such as black holes and the early universe.

- It is a successful framework for describing the behavior of particles and fields at the most fundamental level, incorporating both quantum mechanics and special relativity:

  - At high energies we can witness **particles creation and annihilation**, which cannot be described by non-relativistic quantum mechanics; QFT provides a framework for describing these processes, in an infinite-dimensional system, where the number of particles is not fixed.
  - QFT is a relativistic theory, which means that it is consistent with the principles of special relativity, implmenting **causality** and **Lorentz invariance**; this is essential for describing the behavior of particles at high energies and small distances, where relativistic effects become significant. So two events that are spacelike separated cannot influence each other

    $$(x-y)^2 = (x^0 - y^0) - \sum_{i=1}^{3}(x^i - y^i)^2 < 0 \implies [\mathcal{O}(x), \mathcal{O}(y)] = 0,$$

    thus the events are causally disconnected, ensuring causality.

- Particles are described as excitations of underlying fields, which can be created and annihilated; this allows for a more complete description of particle interactions and the behavior of particles at high energies. Examples of particles and underlying fields include:
  - **Scalar field**: $\phi(x) \implies$ spin-0 particles (e.g., Higgs boson).
  - **Spinor field**: $\psi(x) \implies$ spin-1/2 particles (e.g., electrons, quarks).
  - **Vector field**: $A_\mu(x) \implies$ spin-1 particles (e.g., photons, W and Z bosons).
- The **Standard Model** of particle physics is a quantum field theory that describes the fundamental particles and their interactions (except for gravity); it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. It is a quantum field theory that describes the electromagnetic, weak, and strong interactions, through the exchange of gauge bosons among fields of spin 0, 1/2, and 1 particles.
- The inclusion of gravity in a quantum field theory framework is an open problem in theoretical physics, but can be approached by inserting the metric tensor of general relativity

$$g^{\mu\nu} = \eta^{\mu\nu} + h^{\mu\nu}.$$

  While there are approaches to quantizing gravity, such as string theory and loop quantum gravity, a complete and consistent theory of quantum gravity has not yet been developed. Quanta of the gravitational field are called gravitons, which are hypothetical spin-2 particles that mediate the gravitational interaction, but they have not been observed experimentally. These notes will not cover quantum gravity, but it is an important area of research in theoretical physics which sometimes will be mentioned throughout the notes.

The main goal of these notes is to provide a comprehensive understanding of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level, as well as mathematical methods for making theoretical predictions on interactions among particles. We will cover a wide range of topics, including:

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- The quantization of free fields, which provides the foundation for understanding the behavior of particles and fields in quantum field theory.
- The construction of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level.
- The use of perturbation theory and Feynman diagrams to calculate scattering amplitudes and other physical observables in quantum field theory.
- The renormalization of quantum field theories, which is necessary to make sense of the infinities that arise in perturbation theory and to extract meaningful physical predictions from the theory.
- The application of quantum field theory to the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity).

QFT is one of the most successful and powerful frameworks in physics, and it has been instrumental in our understanding of the fundamental nature of the universe. This theory has been tested and confirmed by a wide range of experiments, some of which tested the predictions of QFT to an extraordinary degree of precision. For example, the anomalous magnetic moment of the electron has been measured to an accuracy of one part in a trillion ($\sim 10^{-13}$), and the predictions of QFT have been confirmed to this level of precision. Additionally, the discovery of the Higgs boson at the Large Hadron Collider in 2012 provided further confirmation of the predictions of QFT and the Standard Model.

Another essential aspect of these notes is to provide a comprehensive understanding of the importance of symmetries in quantum field theory, which play a crucial role in determining the behavior of particles and fields. We will explore the concept of gauge symmetries, which are local symmetries that arise in quantum field theories and are responsible for the interactions between particles and fields. We will also discuss the role of global symmetries, which are symmetries that do not depend on spacetime coordinates, and their implications for the behavior of particles and fields. In the end we will argue that *a QFT is essentially defined by its fields and its symmetries* (statement that will be made more precise in the end of the notes), and we will see how symmetries can be used to constrain the behavior of particles and fields, and to make predictions about their interactions.

For a more comprehensive introduction to quantum field theory, we recommend the following textbooks:

- **An Introduction to Quantum Field Theory** by Michael E. Peskin and Daniel V. Schroeder: This is a widely used textbook that provides a comprehensive introduction to quantum field theory, covering both the theoretical foundations and practical applications of the subject. **TODO:** check and change in the end
- **Quantum Field Theory** by Mark Srednicki: This textbook offers a clear and concise introduction to quantum field theory, with an emphasis on the physical intuition behind the mathematical formalism.
- **Quantum Field Theory in a Nutshell** by A. Zee: This book provides a more informal and intuitive introduction to quantum field theory, making it accessible to a wider audience while still covering essential topics.
- **The Quantum Theory of Fields** by Steven Weinberg: This three-volume series is a comprehensive and rigorous treatment of quantum field theory, written by one of the pioneers in the field. It is suitable for advanced students and researchers who want a deep understanding of the subject.

For more information about the notation and conventions used in these notes, please refer to the appendix, where we provide a detailed explanation of the symbols and mathematical conventions that we will use throughout the notes (Appendix A).

Prerequisites: *Familiarity with the Lagrangian and Hamiltonian formalisms of Classical Mechanics, standard Quantum Mechanics, Special Relativity and the basics of Quantum Field Theory (free fields) is required.*

1 | Free QFT Review

A free quantum field theory does not account for interactions between particles. It describes particles that do not interact with each other, and the fields that represent them evolve independently. Free QFTs are often used as a starting point for understanding more complex interacting theories, as they provide a simpler framework to study the properties of quantum fields and particles.

The lagrangian for a free quantum field theory typically consists of kinetic terms and mass terms for the fields, without any interaction terms: it can only contain **quadratic terms** in the fields, without any higher-order interactions. Quanta of the free fields correspond to particles that do not interact with each other, and their dynamics are governed by the **free propagators** of the theory, which describe how particles propagate through spacetime without any interactions: propagators are the Green's functions of the free theory, and they determine how particles propagate from one point to another in spacetime with determined probability amplitudes.

We will review the free quantum field theories for different types of particles, including scalar fields (spin 0), fermionic fields (spin 1/2), and gauge fields (spin 1). We will also discuss the symmetries of these theories and how they lead to conservation laws through Noether's theorem. This review will provide a foundation for understanding more complex interacting quantum field theories, and it will also highlight the differences between free and interacting theories, as well as the role of symmetries in quantum field theory.

1.1 | Spin 0

For a real scalar field $\phi(x)$, the free Lagrangian is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2, \tag{1.1.1}$$

where m is the mass of the scalar particle. The equation of motion derived from this Lagrangian is the **Klein-Gordon equation**:

$$(\Box + m^2)\phi = 0, \tag{1.1.2}$$

where $\Box = \partial_\mu \partial^\mu$ is the d'Alembertian operator. The solutions to this equation describe free scalar particles with mass m. This is a classical equation of motion, having the same role of Maxwell's equations in classical electromagnetism, but for a scalar field $\phi(x)$ instead of a vector field $A^\mu(x)$.

The idea of quantum field theory is to promote the classical field $\phi(x)$ to a quantum operator $\hat{\phi}(x)$ that acts on a Hilbert space of states. The quantization procedure involves imposing commutation relations on the field operators, which leads to the creation and annihilation of particles. The free scalar field can be expanded in terms of creation and annihilation operators, which allows us to describe the particle content of the theory. Indeed, we can see how decomposing the field into its Fourier modes leads to the interpretation of the quanta of the field as particles with specific momenta and energies:

$$\phi(x) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\, e^{i\mathbf{k}\cdot\mathbf{x}}\tilde{\phi}(t, \mathbf{k}),$$

where each mode can be computed by using this ansatz in the Klein-Gordon equation in momentum space, leading to the dispersion relation of an harmonic oscillator represented by on-shell particles with energy $E_\mathbf{k} = \omega_\mathbf{k} = \sqrt{\mathbf{k}^2 + m^2}$:

$$\left(\partial_t^2 + \omega_\mathbf{k}^2\right)\tilde{\phi}(t, \mathbf{k}) = 0,$$

showing how the free scalar field can be interpreted as a collection of independent harmonic oscillators, each corresponding to a different momentum mode.

In order to quantize the free scalar field, we have to promote the fields themselves to operators and impose the canonical commutation relations, which leads to the creation and annihilation of particles. Before showing how to do that, it is crucial to note that $\hat{\phi}(x)$ depends on both space and time, enforcing the Heisenberg picture of quantum mechanics, where operators evolve in time while states remain fixed. The canonical commutation relations for the scalar field are given by

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}),$$

where $\hat{\pi}(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0\hat{\phi}(x))} = \partial_0\hat{\phi}(x)$ is the conjugate momentum operator. This constraint along with reality of the field $\hat{\phi}(x) = \hat{\phi}^\dagger(x)$ allows us to express the field operator in terms of **creation and annihilation operators**:

$$\hat{\phi}(x) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{k}}}\left(\hat{a}_\mathbf{k}e^{-ik\cdot x} + \hat{a}_\mathbf{k}^\dagger e^{ik\cdot x}\right),$$

where $k \cdot x = \omega_\mathbf{k} t - \mathbf{k}\cdot\mathbf{x} = k^\mu x_\mu$, and $\omega_\mathbf{k} = k^0$. The creation and annihilation (ladder) operators satisfy the following commutation relations

$$\begin{cases} [\hat{a}_\mathbf{k}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3\delta^3(\mathbf{k} - \mathbf{k}'), \\ [\hat{a}_\mathbf{k}, \hat{a}_{\mathbf{k}'}] = [\hat{a}_\mathbf{k}^\dagger, \hat{a}_{\mathbf{k}'}^\dagger] = 0. \end{cases}$$

Note that while the field operator $\hat{\phi}(x)$ is Hermitian and time dependent, the creation and annihilation operators are not, and they are responsible for creating and destroying particles in the quantum field theory. The **vacuum state** $|0\rangle$ is defined as the state that is annihilated by all annihilation operators:

$$\hat{a}_{\mathbf{k}} |0\rangle = 0 \quad \forall \mathbf{k},$$

while we can create one-particle states by acting with the creation operators on the vacuum, which adds one quanta of the field with momentum $\mathbf{k}$:

$$|\mathbf{k}\rangle = \sqrt{2\omega_{\mathbf{k}}} \hat{a}^{\dagger}_{\mathbf{k}} |0\rangle ,$$

and more generally, we can create multi-particle states by acting with multiple creation operators on the vacuum, which leads to the **Fock space** of the theory. With this definition of single particle states, we can compute the inner product between two single particle states, which is given by

$$\langle \mathbf{k} | \mathbf{k}' \rangle = 2\omega_{\mathbf{k}} \delta^3(\mathbf{k} - \mathbf{k}'),$$

showing that the single particle states are orthogonal and normalized with respect to the relativistic normalization convention. The normalization $\sqrt{2\omega_{\mathbf{k}}}$ is arbitrary, but important for ensuring that the probabilities computed in the theory are consistent with relativistic invariance.

Remark. *To find a Lorent invariant measure, we can start from the Minkowski space measure* $\mathrm{d}^4 k$ *and impose the on-shell condition* $k^2 = m^2$ *using a delta function along with a step function to ensure that we only consider positive energy solutions:*

$$\int \mathrm{d}^4 k \, \delta(k^2 - m^2) \theta(k^0) = \int \frac{\mathrm{d}^3 \mathbf{k}}{2\omega_{\mathbf{k}}}.$$

We have then found the Lorentz invariant measure $\frac{\mathrm{d}^3 \mathbf{k}}{2\omega_{\mathbf{k}}}$ *for integrating over the momentum space of on-shell particles, which is crucial for ensuring that the theory is consistent with special relativity. The previous normalization of the single particle states is consistent with this Lorentz invariant measure, since it ensures that the inner product between single particle states is Lorentz invariant, and it also allows us to compute probabilities and cross sections in a way that is consistent with relativistic invariance:*

$$\int \frac{\mathrm{d}^3 \mathbf{k}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}| = 1,$$

which is the completeness relation for the single particle states in the theory. This ensures that we can express any state in the Fock space as a superposition of single particle states, and it also allows us to compute transition amplitudes and probabilities in a way that is consistent with relativistic invariance.

1.1.1 | Feynman's Propagator

1.1.2 | Noether's Theorem

1.2 | Spin 1/2

1.3 | Spin 1

2 Interacting Fields

Free theories are a good starting point to understand the basic concepts and techniques of quantum field theory, but to capture the full complexity of the physical world, we need to introduce interactions between fields. Interactions allow us to describe processes such as scattering, particle decays, and the formation of bound states, which are essential for understanding the behavior of matter and energy at a fundamental level. In a free theory, the fields do not interact with each other, and the equations of motion are linear: they can exhibit *at most quadratic terms in the fields* (in the lagrangian $\mathcal{L}$). We can then build interacting theories by adding non-linear terms to the Lagrangian, which will lead to non-linear equations of motion. These non-linearities are what give rise to the rich phenomenology of interacting quantum field theories.

The most important tools in the developing and understanding of interacting quantum field theories are:

- **Dimensional analysis**, which allows us to determine the relevance of different interaction terms at different energy scales, and to classify interactions as relevant, irrelevant, or marginal.
- **Renormalization**, which is a systematic procedure to deal with the infinities that arise in perturbative calculations of interacting theories, and to extract meaningful physical predictions from them.
- **Symmetry principles**, which play a crucial role in constraining the form of the interactions and in determining the properties of the particles and fields in the theory. For example, gauge symmetries lead to the introduction of gauge fields, which mediate the fundamental forces of nature.

Moreover **locality** is a fundamental principle in quantum field theory, which states that interactions should occur at the same point in spacetime. This principle ensures that the theory respects causality and allows us to construct a consistent framework for describing the dynamics of fields and particles:

$$\mathcal{L}(x) = \mathcal{L}\left(\phi(x), \partial_\mu\phi(x)\right), \quad \mathcal{L}(x) \neq \mathcal{L}\left(\phi(x+y), \partial_\mu\phi(x+y)\right),$$

and the interaction terms we will introduce will be local, meaning that they will involve products of fields and their derivatives evaluated at the same spacetime point x; this ensures that the theory is consistent with the principles of special relativity and quantum mechanics, and it will guide us in constructing the interaction terms in the Lagrangian.

In the next section we will start by performing a dimensional analysis of the fields and the interaction terms, which will help us to understand the behavior of the theory at different energy scales and to identify the most relevant interactions for describing physical phenomena. This will set the stage for our subsequent discussions on renormalization and symmetry principles in interacting quantum field theories.

2.1 | Dimensional Analysis

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2.1.1 | Examples of Interaction Terms

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ϕ^3 Interaction

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ϕ^4 Interaction

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ϕ^n General Interaction

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Complex Scalar Field Interactions

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Quantum Electrodynamics (QED) Interaction

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Yukawa Theory Interaction

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Scalar QED (sQED) Interaction

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2.1.2 | Master Plan

In the next sections, we will introduce interactions in the theory by adding interaction terms to the Lagrangian. The simplest way to do this is to add a term that is a polynomial in the field ϕ, such as $\lambda\phi^4$ or $g\phi^3$. These interaction terms will lead to non-linear equations of motion, which will make the theory more complicated to solve. However, they will also allow us to describe a wider range of physical phenomena, such as scattering processes and particle decays.

More precisely, we are going to:

- Define useful concepts and observables in the theory, such as **cross sections** and **decay rates**.
- Express these observables in terms of the **S-matrix**, which encodes the transition amplitudes between initial and final states in scattering processes: the matrix elements of a generic operator represent the probability amplitude of measuring a specific eigenvalue, corresponding to a transition from an initial state $|i\rangle$ to a final state $|f\rangle$.
- We will use theorems and techniques developed in a QFT framework, such as the **LSZ reduction formula**, to relate the S-matrix elements to correlation functions of the fields or their Fourier transforms, which are the fundamental objects of study in QFT. For example, correlation functions of the form

  $$\langle 0| T\{\phi(x_1)\phi(x_2)\cdots\phi(x_n)\} |0\rangle ,$$

  will show how the fields interact at different points in spacetime, and can be used to compute scattering amplitudes and other physical observables.
- We will develop in the end procedures and techniques to compute these correlation functions from an interacting lagrangian in **perturbation theory**; using **Feynman diagrams** and **Feynman rules** we will be able to systematically calculate the contributions of different interaction terms to the physical observables.

2.2 | Scattering Processes

Scattering processes are currently the most important tool to probe the fundamental interactions of nature, and thus to test our theories of particle physics. They let us infer and understand the properties of particles and their interactions at a microscopic level by analyzing the outcomes of experiments and collisions at macroscopic distances.

We are forced to cite some of the most important results of scattering theory, as well as experiments, which is a vast and complex subject, in order to understand its importance in particle physics. Some of the most important results and experiments in scattering theory include:

- **Rutherford scattering**, which was the first experiment to probe the structure of the atom and led to the discovery of the nucleus. Scattering alpha particles off a thin gold foil, Rutherford observed that most of the particles passed through the foil with little deflection, while a small fraction were scattered at large angles, leading to the conclusion that the atom has a small, dense nucleus at its center. This experiment lead to our modern understanding of the atomic structure of matter at a subatomic level, and it was a crucial step in the development of quantum mechanics and quantum field theory.
- **Lowrence Berkeley National Laboratory**, which was the first particle accelerator and allowed for the discovery of many new particles and interactions: in 1931 Lowrence was the first to accelerate particles to relativistic energies, firstly only around 80 keV, and then with Livingston he built the first cyclotron, which was able to accelerate protons to energies of about 1.22 MeV, leading to the discovery of the neutron in 1932 by James Chadwick.
- **Large Hadron Collider (LHC)**, which is the largest and most powerful particle accelerator in the world, located at CERN, and has been instrumental in the discovery of the Higgs boson in 2012, as well as in probing the properties of the Standard Model and searching for new physics beyond it. It has a circumference of about 27 km and can accelerate protons to energies of up to 7 TeV per beam, allowing for collisions at unprecedented energies and luminosities (actually around 13 TeV, with future upgrades planning further increases).
- **Future colliders**, such as the proposed Future Circular Collider (FCC) at CERN, which aims to reach even higher energies and luminosities than the LHC, and could potentially discover new particles and interactions that are beyond the reach of current experiments. Basically all we know about particle physics and the fundamental interactions of nature has been inferred from the observations and measurements of scattering processes, either in natural phenomena (such as cosmic rays) or in controlled experiments.

Such great energies and luminosities allow us to probe the fundamental interactions of nature at a microscopic level, and to test our theories of particle physics with high precision: the **indetermination principle** allows us to probe smaller and smaller distances by increasing the energy of the particles, thus allowing us to explore the structure of matter and the fundamental forces at a deeper level. By analyzing the outcomes of scattering experiments, we can infer the properties of particles and their interactions, and test our theoretical models against experimental data.

Also the **decay rate** of unstable particles is an important observable that can be measured in experiments and computed in theory, providing insights into the properties of the particles and their interactions. In these decay processes, an unstable particle decays into other smaller and lighter particles, and the decay rate quantifies the probability per unit time that this decay will occur.

By studying the decay rates of particles, we can gain insights into their properties, such as their masses, lifetimes, and interaction strengths, and test our theoretical models against experimental data.

2.2.1 | S-Matrix

We prepare a state $|i,\, t_i\rangle$ at an initial time t_i which will transition after some interaction (or decay, scatter...) into a state $|f,\, t_f\rangle$, measured at a final time t_f with a probability given by the square of the amplitude $|\langle f,\, t_f|i,\, t_i\rangle|^2$. Here we can make the following observations:

- The initial and final states are defined in the limit of infinite past and future, where the particles are free and non-interacting. This is a common assumption in scattering theory, as it allows us to define **asymptotic states** that can be used to compute scattering amplitudes and cross sections.
- The process happens on microscopic time and length scales, while being measured on macroscopic scales, thus we can treat the interaction as an instantaneous event, letting us ignore the detailed time dependencies of the process and focus on the transition between the initial and final states.
- The transition from the initial to the final state can be described by a unitary operator called the **S-matrix**, which encodes the **transition amplitudes** between the asymptotic states.

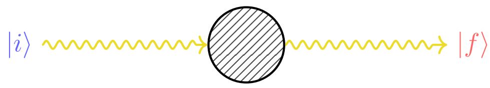


Figure 2.1: Schematic representation of a generic process, where an initial state $|i\rangle$ evolves into a final state $|f\rangle$ through some interaction described by the S-matrix. We are considering asymptotic states, which are defined in the limit of infinite past and future, where the particles are free and non-interacting. The S-matrix encodes the transition amplitudes between these asymptotic states describing the scattering process happening on microscopic time scales.

We tipically use **Heisenberg picture**, so if we want to compute the amplitude of a determined process, we can write it as

$$\langle f|\,\hat{S}\,|i\rangle = \langle f, \infty|i, -\infty\rangle\,,$$

where $|i\rangle$ and $|f\rangle$ are the asymptotic states of the system (time independent and directly related to the free particle states), respectively. *The S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ gives us the amplitude for the transition* from the initial state $|i\rangle$ to the final state $|f\rangle$ due to the interaction described by the S-matrix itself. The S-matrix is a powerful tool in quantum field theory, as it allows us to compute **scattering amplitudes** and **cross sections** for a wide range of processes, and to make predictions that can be tested against experimental data. By analyzing the S-matrix elements for different processes, we can gain insights into the properties of particles and their interactions, testing our theoretical models against experimental observations.

2.3 | Cross Sections

If we analyze a scattering process, such as the Rutherford scattering, we can see how a beam of particles is scattered by a target; in the case of Rutherford scattering, it were alpha particles being scattered by a thin gold foil, where the alpha particles are much smaller than the nucleus of the gold atoms, thus we can treat the nucleus as a still sphere and the alpha particles as point-like particles. We will make the same assumption for the following scattering processes, thus treating the particles as point-like and the interactions as instantaneous events.

In this case, we can define the **cross section** of the target, which quantifies the probability of a scattering event occurring when a particle from the beam interacts with the target. The cross section is an effective area that represents the likelihood of a scattering event, and it can be computed using the S-matrix elements for the process.

The **geometric cross section** of the target, which encodes the effective area of the target for scattering, is given by

$$\sigma \equiv \pi R^2, \tag{2.3.1}$$

where R is the radius of the target. One can compute the number of particles that are scattered by the target, which is given by

$$N = n_B \cdot \sigma, \quad n_B = \frac{N_B}{A},$$

where n_B is the number density of the beam, N_B is the number of particles in the beam, and A is the area of the target. The cross section σ can be interpreted as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target. The power of this approach resides in its simplicity: with few assumptions it allows us to compute the probability of scattering events for a wide range of processes, while linking macroscopical observables, such as the number of scattered particles, to the microscopic dynamics of the interactions encoded in the S-matrix elements.

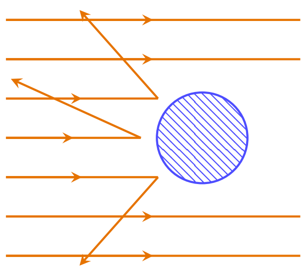


Figure 2.2: Schematic representation of a scattering process, where a beam of particles (orange arrows) is scattered by a target (blue circle). The geometric cross section of the target is given by $\sigma = \pi R^2$, with R being the radius of the target, while the number of scattered particles depends on the interaction probability. We are going to generalize this model to quantum field theory, computing the probability of scattering processes and expressing them in terms of the S-matrix elements.

Now we want to take this model and generalize it to quantum field theory, where we can compute the probability of scattering processes; we want to extend the model

- for any type of scattering process, or interaction;
- for a probability of scattering, not just a number of particles.

2.3.1 | Impact Parameter

If we assume a beam of particles along the z-axis, we can define the **impact parameter b**, which represents the perpendicular distance among the trajectory of the incoming particle and the center of the target:

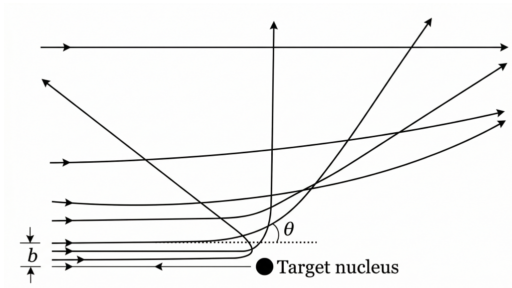
$$N = \int \mathrm{d}^2\mathbf{b}\, n_B\, P(\mathbf{b}) = n_B \int \mathrm{d}^2\mathbf{b}\, P(\mathbf{b}),$$

thus we can now define the **cross section** as

$$\sigma = \int \mathrm{d}^2\mathbf{b}\, P(\mathbf{b}), \tag{2.3.2}$$

where $P(\mathbf{b})$ is the probability of scattering for a given impact parameter $\mathbf{b}$, generalizing the concept of a geometric cross section. This expression allows us to compute the cross section for a scattering process by integrating the probability of scattering over all possible impact parameters, thus providing a more general and flexible framework for analyzing scattering processes in quantum field theory.

Figure 2.3: Schematic representation of a scattering process in terms of the impact parameter $\mathbf{b}$: the probability of scattering can be expressed as a function of the impact parameter, allowing us to compute the cross section by integrating the probability of scattering over all possible impact parameters.



We can also express it in its **differential form**, which ideally counts only a subset of events, with respect to the **solid angle** Ω as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \int \mathrm{d}^2\mathbf{b}\, \frac{P(\mathbf{b})}{\mathrm{d}\Omega}, \tag{2.3.3}$$

where we are considering only the events that are scattered in a specific direction defined by the solid angle $\mathrm{d}\Omega$, thus providing a more detailed description of the scattering process. Alternatively, we can express it with respect to the measure of the space of **final states** $|f\rangle$ as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}f} = \int \mathrm{d}^2\mathbf{b}\, \frac{P(\mathbf{b})}{\mathrm{d}f}. \tag{2.3.4}$$

The consequent step towards the computation of the cross section in QFT are:

- Express the probability of scattering $P(\mathbf{b})$ in terms of the S-matrix elements for the process

  $$\langle f|\,\hat{S}\,|i\rangle = \langle f, \infty | i, -\infty\rangle\,,$$

  which encode the transition amplitudes between the initial and final states.

- Integrate the probability of scattering over all possible impact parameters $\mathbf{b}$ to compute the cross section for the scattering process, using the expressions in equations (2.3.2), (2.3.3), or (2.3.4); then express the result in terms of observables that can be measured in experiments, such as the scattering angle or the energy or momenta of the scattered particles.

2.3.2 | Wave Packets and Regularization

Now one of the natural questions that arises is: what are the initial and final states $|i\rangle$ and $|f\rangle$ that we should use to compute the S-matrix elements for the scattering process? In typical scattering processes, the quantum dispersion of the particle momenta Δp is smaller than the experimental resolution, thus it makes sense to use **eigenstates of the momentum operator**, such as plane wave states. However, there are some issues:

- If the quantum dispersion of the particle momenta Δp is comparable to zero, for the Heisenberg uncertainty principle, the particle is completely delocalized in space

  $$\Delta p \Delta x \geq \frac{\hbar}{2}, \quad \Delta p = 0 \implies \Delta x \to \infty,$$

  thus it can interact with the target at any point in space, leading to divergences in the scattering amplitudes.

- We will show that S-matrix elements for plane wave states are proportional to delta functions, which leads to divergences when we compute the probability of scattering by taking the square of the amplitude: the probability of scattering is given by the square of the amplitude $\langle f| \hat{S} |i\rangle \sim \delta^{(4)}(p_i - p_f)$, but then

  $$P \sim | \langle f| \hat{S} |i\rangle |^2 \sim |\delta^{(4)}(p_i - p_f)|^2 = \delta^{(4)}(p_i - p_f)\delta^{(4)}(0).$$

  The term $\delta^{(4)}(0)$ is divergent, and it arises from the fact that we are using plane wave states, which are not normalizable and lead to divergences in the scattering amplitudes. This is a common issue in quantum field theory, and it requires regularization techniques to handle these divergences and obtain finite results for physical observables.

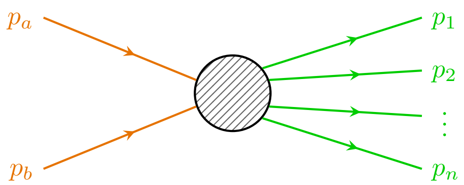
TODO: Add reference to book by schwarz.

One way to overcome the last problem is to consider a finite volume V and a finite time T, so that the delta function can be regularized and then take only at the end the limit $V \to \infty$ and $T \to \infty$. This is a common technique in quantum field theory to handle divergences and make sense of scattering amplitudes. By working in a finite volume and time, we can ensure that the calculations are well-defined and then take the appropriate limits to recover physical results. However, this approach can be obscure and cumbersome, and it can lead to complications in the calculations.

TODO: add reference to Peskin and Schroeder.

We will instead follow the Peskin and Schroeder approach, which is to consider the states as **wave packets** in the momentum space, which are **localized in both position and momentum space**, thus avoiding the issues with the delta function. This approach allows us to compute scattering amplitudes and cross sections without encountering the divergences associated with plane wave states. By using wave packets, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

We can consider a general process in which the initial state is a two particle state, with momenta p_a and p_b, while the final state is a N particle state:



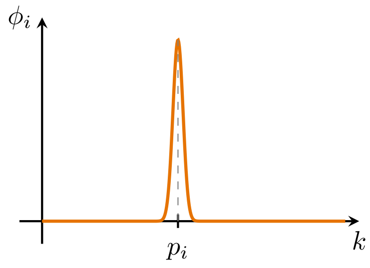
For incoming states we replace $|p_i\rangle$ with $|\phi_i\rangle$, which is a **wave packet** centered around the momentum p_i:

$$|p_i\rangle \to |\phi_i\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \phi_i(\mathbf{k}) \, |\mathbf{k}\rangle \,,$$

with

$$\langle \phi_i | \phi_i \rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} |\phi_i(\mathbf{k})|^2 = 1,$$

remembering that $\langle \mathbf{k}|\mathbf{k}'\rangle = 2E_{\mathbf{k}} \, (2\pi)^3 \delta^{(3)}(\mathbf{k}-\mathbf{k}')$; all of this indicates that $\phi_i(\mathbf{k})$ is a very narrow normalized function centered around p_i.



Example (LHC collider). We can assume, very conservatively, that the LHC collider works with $\Delta p \sim 1\,\mathrm{MeV}$, which is unrealistic (since it would correspond to a resolution of about 10^{-6}, not realistic) and leads to a delocalization of the particles of about

$$\Delta x \sim \frac{\hbar}{2\Delta p} \sim \frac{1}{2\,\mathrm{MeV}} \sim 100\,\mathrm{fm} = 10^{-13}\,\mathrm{m},$$

which, compared to the size of the particles bunches in the LHC, which is of the order of $1\,\mu\mathrm{m} = 10^{-6}\,\mathrm{m}$, is negligible, thus we can safely treat the particles as localized in space, and thus we can use wave packets to compute the scattering amplitudes and cross sections without encountering divergences.

This is a common assumption in high-energy physics, with typically well-localized particles and small quantum dispersion of their momenta compared to the experimental resolution; this allows us to use wave packets to describe the initial and final states in scattering processes.

Now we are left with the task of computing the S-matrix elements for the scattering process using wave packets, and then integrating the probability of scattering over all possible impact parameters to compute the cross section for the scattering process. But how are we going to model the wave packets? We can consider different types of wave packets, such as **Gaussian wave packets**, which are commonly used in quantum mechanics and quantum field theory due to their nice mathematical properties and their ability to represent *localized states in both position and momentum space.* By choosing appropriate wave packet profiles, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

Example (Gaussian wave packet). We can consider a Gaussian wave packet centered around p_i with a width Δp:

$$\phi(\mathbf{k}) = N \exp\left[-\frac{1}{4}\left(\frac{(\mathbf{k}-\mathbf{p})^2}{\Delta p}\right)^2 - i\mathbf{k}\cdot\langle\mathbf{x}\rangle\right], \tag{2.3.5}$$

where N is a normalization constant, Δp is the width of the wave packet in momentum space, and $\langle \mathbf{x}\rangle$ is the average position of the wave packet in real space: using $\hat{x}^\mu = i\frac{\partial}{\partial p_\mu}$, one can check that the uncertainty principle is satisfied with $\Delta x \Delta p = \frac{1}{2}$.

Thus the wave packet is normalized and has a well-defined momentum distribution, with the properties of a momentum eigenstate while being relatively well localized in space for finite Δp. By using Gaussian wave packets, we can compute the S-matrix elements for the scattering process without encountering divergences, and we can ensure that the initial and final states are well-defined and physically meaningful.

Initial states. Sticking to the Gaussian wave packet example, we can write the initial states as the following superposition of momentum eigenstates:

$$|p_a\rangle \to |\phi_a\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{k}}} \phi_a(\mathbf{k})\, |\mathbf{k}\rangle\,,$$
$$|p_b\rangle \to |\phi_b\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{k}}} \phi_b(\mathbf{k})\, e^{-i\mathbf{k}\cdot\mathbf{b}}\, |\mathbf{k}\rangle\,,$$

where we shifted the second wave packet by the impact parameter $\mathbf{b}$, defined as

$$\mathbf{b} = \langle\phi_b|\, \hat{\mathbf{x}}^\perp \,|\phi_b\rangle\,,$$

with $\hat{\mathbf{x}}^\perp$ being the position operator in the plane perpendicular to the beam axis. Looking at the expression for the gaussian wave packet in (2.3.5), we can think of having took out of ϕ_b the exponential factor $e^{-i\mathbf{k}\cdot\langle\hat{\mathbf{x}}^\perp\rangle}$, whereas in ϕ_a it can be set to zero by choosing the average position of the wave packet to be at the origin. In this way we can model the initial states as wave packets that are localized in both position and momentum space, with the second wave packet shifted by the impact parameter $\mathbf{b}$ with respect to the first one, to account for the spatial separation of the incoming particles in the scattering process.

Outgoing states. For the outgoing states we can consider just the momentum eigenstates, since the scattering has already happened, thus we can write

$$|\phi_i\rangle = |p_i\rangle\,, \quad \forall i = 1, \ldots, N.$$

Since the outgoing particles are detected at macroscopic distances, we can treat them as free particles and thus we can use momentum eigenstates to describe them: the scattering already happened, thus we can treat the outgoing particles as free particles.

Now we can compute the **S-matrix elements** for the scattering process using the initial and final states defined above. We can write the S-matrix elements for momentum eigenstates as

$$\langle f,\, \infty | i,\, -\infty\rangle = \langle f|\, \hat{S}\, |i\rangle = \langle p_1, \ldots,\, p_N|\, \hat{S}\, |p_a, p_b\rangle\,,$$

where we are considering the initial state as a two particle state with momenta p_a and p_b, and the final state as a N particle state with momenta $p_1, \ldots, p_N$. The S-matrix can be expressed as

$$\hat{S} = \mathbb{I} + i\hat{T},$$

where the identity term corresponds to the case where no scattering occurs, while the $i\hat{T}$ term corresponds to the case some interactions among the fields, i.e. the scattering occurs. Thus we can write[1]

$$\langle p_1, \ldots,\, p_N|\, i\hat{T}\, |p_a, p_b\rangle = (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_{i=1}^{N} p_i) i\mathcal{M},$$

where the delta function ensures the conservation of energy and momentum, and $\mathcal{M}$ is the **invariant matrix element** or **scattering amplitude** for the process, which encodes the dynamics

[1]Note that we evaluate the transition amplitude here using exact momentum eigenstates $|p_a, p_b\rangle$ rather than the initial wave packets $|\phi_a, \phi_b\rangle$. This is done to formally define the invariant scattering amplitude $\mathcal{M}$ and cleanly factor out the energy-momentum conserving Dirac delta $\delta^{(4)}$. The physical wave packets enter the calculation in the subsequent step: exploiting the linearity of the $\hat{T}$ operator, the realistic transition amplitude is obtained by integrating these exact-momentum matrix elements over the momentum distributions $\phi_a(\mathbf{k})$ and $\phi_b(\mathbf{k})$ of the incoming wave packets.

of the interaction (we omitted its dependencies for simplicity, but it relies on the initial and final states, thus $\mathcal{M} = \mathcal{M}(p_a, p_b \to p_1, \ldots, p_N)$). The invariant amplitude can be computed using Feynman diagrams and the rules of quantum field theory, and it is a key quantity in determining the cross section for the scattering process.

Thus we can now compute the **differential cross section** in the space of final configurations as in (2.3.4):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}f} = \int \mathrm{d}^2\mathbf{b}\, \frac{\mathrm{d}P}{\mathrm{d}f},$$

where the differential probability of scattering in the space of final states is given by

$$\frac{\mathrm{d}P}{\mathrm{d}f} = |\left\langle p_1,\, \ldots,\, p_N,\, +\infty\right| i\hat{T} \left|\phi_a,\, \phi_b,\, -\infty\right\rangle|^2,$$

and the measure of the space of final states for N particles is normalized as[2]

$$\mathrm{d}f = \prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{p}_i}{(2\pi)^3} \frac{1}{2E_{p_i}}.$$

Now we can put everything back together and after some algebra (full computation in Appendix B) we can write the final expression for the differential cross section **TODO:** write it.

$$\begin{aligned}
\mathrm{d}\sigma &= \frac{1}{\Phi} |\mathcal{M}(p_a, p_b \to p_1, \ldots, p_N)|^2 \mathrm{d}\Pi_N, \\
\mathrm{d}\Pi_N &= \left[\prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{p}_i}{(2\pi)^3 2E_{p_i}}\right] (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_{i=1}^{N} p_i), \\
\Phi &= 2E_a 2E_b |v_a - v_b| = 4|E_b p_a^z - E_a p_b^z|,
\end{aligned}$$

where:

- $\mathrm{d}\Pi_N$ is the Lorentz-invariant phase space measure for N final state particles, which accounts for the density of final states and the conservation of energy and momentum through the delta function.
- Φ is the flux factor, which accounts for the incoming particles and their relative velocity, ensuring that the cross section is properly normalized with respect to the initial state of the scattering process.
- $\mathcal{M}$ is again the invariant matrix element or scattering amplitude, which encodes the dynamics of the interaction and can be computed using Feynman diagrams and the rules of quantum field theory.

Φ is the only non Lorentz invariant term: we have indeed

$$\Phi = 4|E_b p_a^z - E_a p_b^z|,$$

where z is the beam axis, thus it is not invariant. But we can recognize its invariance under:

[2] We can convince ourself with the resolution of the identity for 1 particle states, which reads

$$\mathbb{I} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{2E_\mathbf{k}} \left|\mathbf{k}\right\rangle \left\langle\mathbf{k}\right|,$$

where $\mathrm{d}f = \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{2E_\mathbf{k}}$ is the measure of the space of final states for 1 particle.

1. boosts along the beam axis, since E_a and p_a^z transform in the same way, and E_b and p_b^z also transform in the same way, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under boosts along the beam axis.

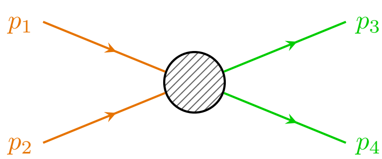
2. rotations around the beam axis (in the perpendicular plane), since E_a, E_b, p_a^z and p_b^z are scalars under rotations around the beam axis, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under rotations around the beam axis.

Thus the flux factor Φ transforms as the inverse of an area in the plane perpendicular to the beam axis, behaving as a cross section (which is consistent with the interpretation of the cross section as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target).

Even if we focus on the case of a beam of particles scattering on a target, the computed expression for the differential cross section remains symmetric for exchange of beam and target, thus it can be applied to any scattering process, regardless of the specific roles of the particles involved. We can also imagine to use it for collisions of two beams of particles, as in e^+e^- collisions in LEP, $p\bar{p}$ collisions in the Tevatron, or more recently pp collisions in the LHC.

We are basically performing a change of reference frame in order to reverse the problem: we can consider the target as the beam and the beam as the target, it all depends on the symmetries and the choice of the reference frame. This is a common technique in scattering theory, where we can choose a reference frame that simplifies the calculations (even center of mass, it depends) and makes it easier to analyze the scattering process. By considering different reference frames, we can gain insights into the dynamics of the interaction and compute cross sections more efficiently.

Example (2 → 2 scattering in center of mass frame). We can consider a simple scattering process where two particles collide and produce two particles in the final state. In the center of mass frame, the initial momenta of the particles are equal and opposite, and we can analyze the scattering process using the invariant amplitude $\mathcal{M}$ and the phase space factor Φ to compute the differential cross section for this process.



The incoming particles have momenta p_1 and p_2, and the outgoing particles have momenta p_3 and p_4, and their spatial parts is conserved: $\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{p}_3 + \mathbf{p}_4 = 0$ (since we are in the center of mass frame). The energy of the particles is given by the energy momentum relation $E_i = \sqrt{\mathbf{p}_i^2 + m_i^2}$, where m_i is the mass of the particle. We can write

$$\mathrm{d}\Pi_2 = \frac{\mathrm{d}^3\mathbf{p}_3}{(2\pi)^3 2E_3} \frac{\mathrm{d}^3\mathbf{p}_4}{(2\pi)^3 2E_4} (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4),$$

which, after integrating in $\mathrm{d}^3\mathbf{p}_4$ using the delta function, gives $\mathbf{p}_4 = -\mathbf{p}_3$; using then polar coordinates for $\mathrm{d}^3\mathbf{p}_3 = p^2\,\mathrm{d}p\,\mathrm{d}\Omega$, we can write

$$E_3 = \sqrt{p^2 + m_3^2}, \quad E_4 = \sqrt{p^2 + m_4^2}, \quad p = |\mathbf{p}_3| = |\mathbf{p}_4|,$$

and thus

$$\int \mathrm{d}p\, \delta(E_1 + E_2 - E_3 - E_4) = \left| \frac{p}{E_3} + \frac{p}{E_4} \right|^{-1} = \frac{E_3 E_4}{p(E_3 + E_4)} = \frac{E_3 E_4}{pE},$$

where the last energy E is the total energy of the system in the center of mass frame, which is given by $E = E_1 + E_2 = E_3 + E_4$. Thus we can write

$$\mathrm{d}\Pi_2 = \mathrm{d}\Omega \frac{p}{16\pi^2 E},$$

with a flux $\Phi = 2E_1 2E_2 |v_1 - v_2|$, with now

$$|v_1 - v_2| = \left| \frac{|\mathbf{p}_1|}{E_1} + \frac{|\mathbf{p}_2|}{E_2} \right| = p_{in} \frac{E}{E_1\, E_2},$$

where p_{in} is the magnitude of the incoming momenta (which are equal in the center of mass frame) $p_{in} = |\mathbf{p}_1| = |\mathbf{p}_2|$. Thus we can write the final expression for the differential cross section as

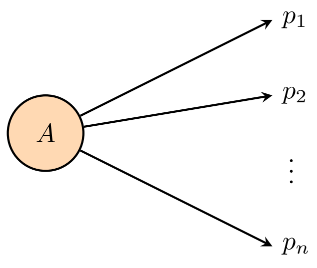
$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{64\pi^2 E^2} \frac{p}{p_{in}} |\mathcal{M}|^2,$$

where we used the fact that $E_1 E_2 = \frac{E^2}{4}$ in the center of mass frame. This expression allows us to compute the *differential cross section for the 2 → 2 scattering process only in the center of mass frame*, given the invariant amplitude $\mathcal{M}$ and the kinematic variables of the process.

> **Remark.** *Most of the time this problem is independent of the azimuthal angle* ϕ*, thus we can integrate over it and write* $\mathrm{d}\Omega = 2\pi \mathrm{d}\cos\theta$*, thus simplifying the expression for the differential cross section.*
>
> *Furthermore* $\mathrm{d}\sigma$ *is invariant under boosts along the beam axis, but the solid angle* $\mathrm{d}\Omega$ *is not, thus the expression for the differential cross section* $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ *is not invariant.*

2.3.3 | Decay Rates

We can similarly deal with **unstable particles**, which after a certain lifetime can decay into other lighter particles. Systems composed by unstable particles can be analyzed using the S-matrix formalism in a similar way to how we compute scattering amplitudes and cross sections for stable particles. By analyzing the transition amplitudes for the decay process, we can determine the decay rate and the branching ratios for different decay channels, which provide insights into the properties of the unstable particle and its interactions with other particles.



The **decay rate** Γ of an unstable particle can be computed as

$$\Gamma = \frac{\text{Number of decays per unit time}}{\text{Number of particles}},$$

and we can define the **lifetime** τ of the particle as the inverse of the decay rate:

$$\tau = \frac{1}{\Gamma}.$$

Decay rates and lifetimes are defined in the rest frame of the particle, where the particle is at rest and its energy is equal to its mass $E = m$ and its momentum is zero $\mathbf{p} = 0$.

The decay rate can be computed using the S-matrix element for the transition from the initial state (the unstable particle) to the final state (the decay products), and it is related to the **invariant**

amplitude $\mathcal{M}$ for the decay process. We cannot directly apply the limit for $t_i \to -\infty$, since the particle is unstable and it would have decayed before, it cannot exist for an infinite amount of time. We therefore need to consider a finite time interval for the decay process, but we can make a simplification: if we assume the particle to have a lifetime way longer than the time scale of the interaction (always true for most practical cases), we can still apply the S-matrix formalism and compute the decay rate using the invariant amplitude for the decay process, and we will have to take into account the instability of the particle only in one step of the calculation.[3]

TODO: check footnote.

Defining the **differential decay rate** as

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}f} = \frac{1}{T}\mathrm{d}P, \tag{2.3.6}$$

where T is the time interval during which the decay process occurs $T = t_f - t_i$, and $\mathrm{d}P$ is the probability of decaying from the initial state to the final state within that time interval; we can compute the total decay rate by integrating over the phase space of the final states; let us first write the expression for the delta function in the limit of finite and large T:

$$\delta(0) = \int_{t_i}^{t_f} \mathrm{d}t\, e^{i(E_i - E_f)t}|_{t_f\to\infty, t_i\to-\infty},$$

which is the critical step in the calculation, since it allows us to regularize the delta function and make sense of the transition amplitude for the decay process; the complete calculation is shown in Appendix B. Having solved the subtleties of defining the asymptotic states for unstable particles, the final expression for the decay rate after integrating over the phase space of the final states is

TODO: write it.

$$\mathrm{d}\Gamma = \frac{1}{2m}|\mathcal{M}_{p_i\to p_f}|^2 \mathrm{d}\Pi_N, \tag{2.3.7}$$

where m is the mass of the unstable particle, $\mathcal{M}_{p_i\to p_f}$ is the invariant amplitude for the decay process, and $\mathrm{d}\Pi_N$ is the phase space factor for the N-particle final state, which accounts for the kinematics of the decay products and ensures that energy and momentum are conserved in the decay process. It is very similar to the expression for the differential cross section, with some differences concerning the flux factor term $\Phi = 2E_a 2E_b|v_a - v_b|$:

- $2E_a \to 2m$ since we are in the rest frame of the unstable particle, thus $E_a = m$;
- $2E_b|v_a - v_b|$ is not present since we are not dealing with a scattering process, but rather with a decay process, thus there is no incoming particle b and no relative velocity to consider.

This expression allows us to compute the decay rate for an unstable particle given the invariant amplitude and the kinematic variables of the decay process.

Identical particles. When dealing with identical particles among the decay products, the previous formula are still valid, with the appropriate modifications to take into account the symmetrization (for bosons) or antisymmetrization (for fermions) of the wave function. This means that the scattering amplitude must be modified to include the appropriate symmetry factors, which can affect the calculation of cross sections and decay rates in order to consider the spin statistics theorem

$$|p_3,\, p_4\rangle = \pm\, |p_4,\, p_3\rangle\,.$$

[3] We will consider a finite time interval for the decay process and then take the limit of large time intervals to recover the usual expression for the decay rate.

This can be accomplished in two ways: one can restrict the integration over the phase space to a specific region that accounts for the indistinguishability of the particles (thus over all final inequivalent states), or one can include a symmetry factor in the calculation of the amplitude (such as the permutation number $\frac{1}{N!}$) to account for the number of identical particles in the final state. Both approaches ensure that the physical predictions are consistent with the principles of quantum mechanics and the statistics of identical particles.

2.4 | LSZ reduction and Green's functions

In this section we aim to show how the S-matrix elements can be computed from the Green's functions of the interacting theory. This is done by the LSZ reduction formula, which relates the S-matrix elements to the time-ordered correlation functions of the fields.

The Green's functions of the interacting theory are defined as the time-ordered correlation functions of the field operators in the interacting theory, evaluated in the vacuum state of the interacting theory:

$$\langle f| \hat{S} |i\rangle \to \langle\Omega| T\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\} |\Omega\rangle \tag{2.4.1}$$

where

- $\hat{\phi}$ is the field operator of the interacting theory,
- $|\Omega\rangle$ is the vacuum state of the interacting theory,

where we highlighted the difference among the vacuum state of the interacting theory and the free theory one: interactions can modify the structure of the vacuum and the Green's functions of the interacting theory encode all the information about its dynamics; they are the building blocks for computing physical observables such as scattering amplitudes and cross sections.

We now start from the structure analysis of the two point functions $\langle\Omega| \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} |\Omega\rangle$ in an interacting theory, investigating its physical content and how it relates to the S-matrix elements $\langle f| \hat{S} |i\rangle$, finally relating it to the LSZ reduction formula for connecting the scattering matrix elements to the n-point Green's functions. We will start from a **real scalar theory** before generalizing to more complicated theories.

2.4.1 | Two Point Functions

TODO: Add reference to Peskin 7.1

To analyze the structure of the two point function, we start from the definition of the time-ordered two point function in the interacting theory:

$$\langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(y)\} |\Omega\rangle ,$$

the idea is now to insert a complete set of physical states $\langle\lambda|$[4] as a resolution of the identity $\mathbb{I} = \sum_\lambda |\lambda\rangle\langle\lambda|$ between the two field operators; but we have to be more precise, since the vacuum state $|\Omega\rangle$ is a physical state of the theory, we have to separate it from the other states in the spectrum of the theory, so that we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle\langle\Omega| + \sum_{\lambda\neq\Omega} |\lambda\rangle\langle\lambda| ,$$

where $|\lambda\rangle$ are seen as eigenstates of the momentum operator $\hat{\mathbf{P}}$ plus other quantum numbers, so that we can write them as $|\lambda_{\mathbf{p}}\rangle$ to make the momentum dependence explicit. They are normalized according to the relativistic normalization convention, so that we have

$$\langle\lambda_{\mathbf{p}}|\lambda'_{\mathbf{p}'}\rangle = (2\pi)^3 2E_{\mathbf{p}}(\lambda)\delta^{(3)}(\mathbf{p}-\mathbf{p}')\delta_{\lambda,\lambda'},$$

[4] Here we can interpret λ as the set of quantum numbers characterizing the different states of our theory.

where $E_{\mathbf{p}}(\lambda) = \sqrt{\mathbf{p}^2 + m_\lambda^2}$ is the energy of the state $|\lambda_{\mathbf{p}}\rangle$ with momentum $\mathbf{p}$ and mass m_λ. The sum over λ includes all the physical states of the theory, such as the one particle states, the two particle states, the bound states, etc.

We cited the mass m_λ of the state $|\lambda_{\mathbf{p}}\rangle$ as a label for the state, but it is important to note that the mass m_λ is not a free parameter, but it is determined by the dynamics of the theory and it is related to the eigenvalue of the momentum operator $\hat{P}^\mu$ on the state $|\lambda_{\mathbf{p}}\rangle$ by the relativistic dispersion relation:

$$\hat{P}^\mu |\lambda_{\mathbf{p}}\rangle = p^\mu |\lambda_{\mathbf{p}}\rangle, \quad p^2 = p_\mu p^\mu = m_\lambda^2,$$

so that the mass m_λ is determined by the eigenvalue of $\hat{P}^2$ on the state $|\lambda_{\mathbf{p}}\rangle$:

$$\hat{P}^2 |\lambda_{\mathbf{p}}\rangle = m_\lambda^2 |\lambda_{\mathbf{p}}\rangle .$$

Putting everything together, we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle \langle\Omega| + \sum_{\lambda\neq\Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| ,$$

with the factor of $1/(2E_{\mathbf{p}}(\lambda))$ coming from the relativistic normalization of the states $|\lambda_{\mathbf{p}}\rangle$.

If we now assume $x^0 > y^0$, the time-ordering operator T does not change the order of the fields, and we can neglect it in order to write the two point function as:

$$\begin{aligned}
\langle\Omega| \hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle &= \langle\Omega| \hat{\phi}(x) \left(|\Omega\rangle \langle\Omega| + \sum_{\lambda\neq\Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \right) \hat{\phi}(y) |\Omega\rangle \\
&= \langle\Omega| \hat{\phi}(x) |\Omega\rangle \langle\Omega| \hat{\phi}(y) |\Omega\rangle + \sum_{\lambda\neq\Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} \langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \hat{\phi}(y) |\Omega\rangle ,
\end{aligned}$$

and if we now relate $\hat{\phi}(x)$ to $\hat{\phi}(0)$ by a spacetime translation, using the unitary operator $\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x}$ that implements the translation, so that

$$\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x} \implies \hat{\phi}(x) = e^{i\hat{P}\cdot x}\, \hat{\phi}(0)\, e^{-i\hat{P}\cdot x} = \hat{U}_{\mathbf{p}}\, \hat{\phi}(0)\, \hat{U}_{\mathbf{p}}^\dagger,$$

we can compute the terms by assuming that the vacuum is *translationally invariant*, and since we said that $|\lambda_{\mathbf{p}}\rangle$ are eigenstates of the momentum operator $\hat{P}$ with eigenvalue p, we can write:

$$\begin{aligned}
\langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle &= \langle\Omega| e^{i\hat{P}\cdot x}\hat{\phi}(0)e^{-i\hat{P}\cdot x} |\lambda_{\mathbf{p}}\rangle \\
&= e^{-ip\cdot x} \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle ,
\end{aligned}$$

from vacuum invariance and momentum eigenstate properties. We can now assume a Lorentz transformation Λ and the corresponding unitary operator $\hat{U}(\Lambda)$ to exist such that $\Lambda p = (m_\lambda, \mathbf{0})$ and $\hat{U}(\Lambda)|\lambda_{\mathbf{p}}\rangle = |\lambda_{\mathbf{0}}\rangle$. Hence, inserting the identity operator $\hat{U}(\Lambda)^\dagger\hat{U}(\Lambda) = \mathbb{I}$ two times, we can write

$$\langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle = \left(\langle\Omega| \hat{U}(\Lambda)^\dagger\right) \left(\hat{U}(\Lambda)\hat{\phi}(0)\hat{U}(\Lambda)^\dagger\right) \left(\hat{U}(\Lambda) |\lambda_{\mathbf{p}}\rangle\right) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle ,$$

since the vacuum is invariant under Lorentz transformations, and the field operator transforms trivially: $\hat{\phi}(\Lambda 0) = \hat{\phi}(0)$. Thus we can write

$$\langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle\, e^{-ip\cdot x} = e^{-ip\cdot x} F(\lambda),$$

where we have defined the function $F(\lambda) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle$ which encodes the information about the coupling of the field operator to the state $|\lambda\rangle$. In the particular case of $|\lambda_{\mathbf{p}}\rangle = |\Omega\rangle$, we get

$$\langle\Omega| \hat{\phi}(x) |\Omega\rangle = v = \text{constant},$$

which is a consequence of the translational invariance of the vacuum state and represents the vacuum expectation value of the field operator. We can then always redefine the field operator by subtracting the vacuum expectation value

$$\phi(x) = v + \tilde{\phi}(x), \quad \langle\Omega|\, \tilde{\phi}(x)\, |\Omega\rangle = 0,$$

since the vacuum expectation value of the field operator does not affect the dynamics of the theory, and it can be absorbed into a redefinition of the field operator. This means that we can always choose a field operator such that its vacuum expectation value is zero, which simplifies the analysis of the two point function and the LSZ reduction formula. Thus the two point (after the redefinition of the field operator) function can be written as

$$\langle\Omega|\, \hat{\phi}(x)\hat{\phi}(y)\, |\Omega\rangle = \sum_\lambda |F(\lambda)|^2 \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} e^{-ip\cdot(x-y)},$$

valid for $x^0 > y^0$ and where the last term

$$D(x,\, y,\, m_\lambda) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} e^{-ip\cdot(x-y)} = \langle 0|\, \hat{\phi}_{0,\lambda}(x)\hat{\phi}_{0,\lambda}(y)\, |0\rangle\,, \tag{2.4.2}$$

is the **free propagator** of a particle with mass m_λ, with $\hat{\phi}_{0,\lambda}$ the free field operator of a particle with mass m_λ: it is the amplitude for a free scalar particle to propagate from point y to point x. We can now reintroduce the time-ordering to write the two point function in a more general form

$$\begin{aligned}\langle\Omega|\, T\{\hat{\phi}(x)\hat{\phi}(y)\}\, |\Omega\rangle &= \Theta(x^0 - y^0)\, \langle\Omega|\, \hat{\phi}(x)\hat{\phi}(y)\, |\Omega\rangle + \Theta(y^0 - x^0)\, \langle\Omega|\, \hat{\phi}(y)\hat{\phi}(x)\, |\Omega\rangle \\ &= \Theta(x^0 - y^0) \sum_\lambda |F(\lambda)|^2 D(x,\, y,\, m_\lambda) + \Theta(y^0 - x^0) \sum_\lambda |F(\lambda)|^2 D(y,\, x,\, m_\lambda).\end{aligned}$$

In this last expression we can recognize the **Feynman propagator** of a particle with mass m_λ:

$$\begin{aligned}D_F(x - y,\, m_\lambda) &= \Theta(x^0 - y^0) D(x,\, y,\, m_\lambda) + \Theta(y^0 - x^0) D(y,\, x,\, m_\lambda) \\ &= \langle 0|\, \hat{T}\{\hat{\phi}_{0,\lambda}(x)\hat{\phi}_{0,\lambda}(y)\}\, |0\rangle = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{i}{p^2 - m_\lambda^2 + i\epsilon} e^{-ip\cdot(x-y)},\end{aligned} \tag{2.4.3}$$

which is the amplitude for a free scalar particle to propagate from point y to point x in a time-ordered manner, and it is given by the Fourier transform of the free propagator with the appropriate $i\epsilon = i0^+$ prescription to ensure causality.

With this last definition, we can now write the final expression, the **Källén-Lehmann spectral representation** of the two point function, as:

$$\langle\Omega|\, T\{\hat{\phi}(x)\hat{\phi}(y)\}\, |\Omega\rangle = \int \frac{\mathrm{d}M}{2\pi} g(M^2) D_F(x,\, y,\, M), \tag{2.4.4}$$

where we have defined the **spectral density** function $g(M^2)$ as

$$g(M^2) = \sum_\lambda (2\pi) |F(\lambda)|^2 \delta(M^2 - m_\lambda^2). \tag{2.4.5}$$

The spectral density function encodes the information about the spectrum of the theory,[5] and it is a positive function since it is defined as a sum of positive terms. The Källén-Lehmann representation of the two point function shows that the two point function can be written as a superposition of free propagators with different masses, weighted by the spectral density function.

[5]The spectral density function is a measure of the number of states with a given mass: the spectrum of the theory is the set of all possible masses of physical particles.

This spectral representation is very useful for analyzing the properties of the theory, and it will be crucial for deriving the LSZ reduction formula in the next section: it will allow us to relate the S-matrix elements to the poles of the two point function (which correspond to the physical particles in the spectrum of the theory) and to the residues of these poles (which correspond to the field strength renormalization constants of these particles).

Let us make some observations about the Källén-Lehmann representation to understand better the look and physical meaning of the spectral density function $g(M^2)$:

- If $M^2 = m^2$ with m the physical mass of a **one particle state** (i.e. m^2 eigenvalue of $\hat{P}^2$), then the spectral density function has a delta function contribution

$$g(M^2)\Big|_{M^2=m^2} = Z(2\pi)\delta(M^2 - m^2),$$

with $Z = |F(\lambda = 1\text{-part. state})|^2$ which represents the *field strength renormalization constant* of the one particle state. This means that the two point function has a pole at $p^2 = m^2$ with residue Z, which is a consequence of the presence of a one particle state in the spectrum of the theory.

- For a **two particle bound state** we have two cases:
  - If $M^2 \leq (2m)^2$ then the spectral density function has a delta function contribution

$$g(M^2)\Big|_{M^2\leq(2m)^2} = \sum_{\lambda\in 2\text{-part. states}} Z_B(2\pi)\delta(M^2 - m_\lambda^2),$$

  with $Z_B = |F(\lambda = 2\text{-part. state})|^2$ which represents the *field strength renormalization constant* of the two particle bound state. This means that the two point function has a pole at $p^2 = M_B^2$ with residue Z_B, which is a consequence of the presence of a two particle bound state in the spectrum of the theory.
  - If $M^2 > (2m)^2$ then the spectral density function has a continuous contribution given by the two particle continuum states, which represent the scattering states of two particles with mass m

$$g(M^2) = Z_C \sum_{\lambda\in 2\text{-part. cont. states}} (2\pi)\delta(M^2 - m_\lambda^2),$$

  with $Z_C = |F(\lambda = 2\text{-part. cont. state})|^2$ which represents the *field strength renormalization constant* of the two particle continuum states. This means that the two point function has a branch cut starting at $p^2 = (2m)^2$, which is a consequence of the presence of the two particle continuum states in the spectrum of the theory.

- The spectral density function $g(M^2)$ is a positive function of the **bare mass** M, since it is defined as a sum of positive terms. This means that the two point function is a positive function, which is a consequence of the unitarity of the theory.

- The physical mass m^2 is the eigenvalue of $\hat{P}^2$, so that it represents the mass of the one particle state in the spectrum of the theory. The physical mass is different from the bare mass M^2 which appears in the Lagrangian of the theory, and it is related to the bare mass by a renormalization procedure.

Thus we can now understand a typical look of the spectral density function $g(M^2)$ for a theory with a one particle state, some two particle bound states and a two particle continuum states:

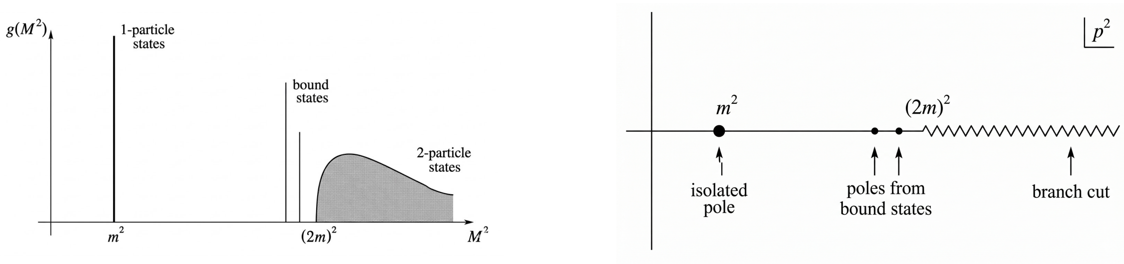


Figure 2.4: On the left image we can see the spectral density function for a theory with a one particle state and a two particle bound and continuum states, with the delta function contributions and the continuous part. On the right image we can see the analytic structure in the complex p^2-plane of the two point function Fourier transform, with the pole at $p^2 = m^2$ corresponding to the one particle state, the poles at $p^2 = M_B^2 \leq (2m)^2$ corresponding to the two particle bound states, and the branch cut starting at $p^2 = (2m)^2$ corresponding to the two particle continuum states.

We can now study again the two point function in momentum space, by performing a Fourier transform of the Källén-Lehmann representation:

$$\begin{aligned}\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(0)\} |\Omega\rangle &= \int \mathrm{d}^4x\, e^{ip\cdot x} \int \frac{\mathrm{d}M}{2\pi} g(M^2) D_F(x,\, 0,\, M) \\ &= \frac{iZ}{p^2 - m^2 + i0^+} + \int_{\sim(2m)^2}^{\infty} \frac{\mathrm{d}M^2}{2\pi} \frac{ig(M^2)}{p^2 - M^2 + i\epsilon},\end{aligned}$$

where we have divided the integral over M^2 into the contribution from the one particle state, which gives a pole at $p^2 = m^2$ with residue iZ, and the contribution from the two particle states, which gives a branch cut starting at $p^2 = (2m)^2$ (these are regular terms for $p^2 \sim m^2$). The contribution from the two particle bound states, if present, would give poles at $p^2 = M_B^2 \leq (2m)^2$ with residue iZ_B (as shown in the above figure), but we have not explicitly written them in the last expression for simplicity. To sum up, in momentum space the two point function has the following analytic structure in the complex p^2-plane:

1. A branch cut starting at $p^2 = (2m)^2$ due to the presence of the two particle continuum states in the spectrum of the theory, which represent the scattering states of two particles with mass m.

2. A pole at $p^2 = m^2$ with residue iZ, which is a consequence of the presence of a one particle state in the spectrum of the theory.

3. Poles at $p^2 \leq (2m)^2$ for each two particle bound state in the spectrum of the theory, with residue iZ_B, which is a consequence of the presence of the two particle bound states in the spectrum of the theory.

For a free theory we would have only the one particle state in the spectrum of the theory, so that the Källén-Lehmann representation of the two point function would reduce to

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle 0| T\{\hat{\phi}_0(x)\hat{\phi}_0(0)\} |0\rangle = \frac{i}{p^2 - m^2 + i0^+},$$

with $Z = 1$ since the field operator creates a one particle state with unit probability, and the physical mass m^2 coincides with the bare mass M^2 since there are no interactions in the theory.

For an interacting theory we will have corrections to the physical mass and to the field strength renormalization constant, so that we can write

$$\begin{cases} m = m_\lambda + O(g), \\ Z = 1 + O(g), \end{cases}$$

where g is the coupling constant of the theory, so that the physical mass m^2 and the field strength renormalization constant Z receive corrections from the interactions in the theory.

Example (Dirac Field). In the end we will give rules for the generalization of the LSZ reduction formula to more complicated theories, such as theories with fermions and gauge fields, but let us first analyze the structure of the two point function for a Dirac field. If we compute these last results for a Dirac field, we would find

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| \hat{T}\{\hat{\psi}(x)\hat{\bar{\psi}}(0)\} |\Omega\rangle = \frac{iZ(\not{p}+m)}{p^2 - m^2 + i\epsilon},$$

such that

$$\langle\Omega| \hat{\psi}(0) |\lambda_{\mathbf{p},s}\rangle \Big|_{\lambda=1\text{-part}} = u_s(\mathbf{p}) F(\lambda = 1\text{-part}) = u_s(\mathbf{p})\sqrt{Z},$$

where $u_s(\mathbf{p})$ is the Dirac spinor for a particle with momentum $\mathbf{p}$ and spin s, with

$$(\not{p}+m) = \sum_s u_s(\mathbf{p})\overline{u}_s(\mathbf{p}),$$

which is the completeness relation for the Dirac spinors. The spectral density function for the Dirac field would have a delta function contribution at $M^2 = m^2$ with residue Z, which corresponds to the one particle state in the spectrum of the theory, and a branch cut starting at $M^2 = (2m)^2$ due to the presence of the two particle continuum states in the spectrum of the theory, which represent the scattering states of two particles with mass m. We would not have two particle bound states in the spectrum of the theory, since the Dirac field does not have self-interactions, so that the spectral density function would not have delta function contributions at $M^2 \leq (2m)^2$ corresponding to two particle bound states. **(?)** Is this last statement true?

Example (Unstable Particle State). If we are considering unstable particles, i.e particles with a decay rate $\Gamma \neq 0$, then the last results are not valid anymore, since we adopted eigenvalues of the momentum operator $\hat{P}^\mu$, and in particular of $\hat{H}$, which cannot be defined for unstable particles:

$$\hat{H} |\psi\rangle = E |\psi\rangle, \hat{S}(t,\, t_0) |\psi\rangle = e^{i\hat{H}(t-t_0)} |\psi\rangle = e^{iE(t-t_0)} |\psi\rangle,$$

since an eigenstate by itself can only get a phase and does not decay, while an unstable particle state has to decay in order to be unstable.

Thus a rigorous definition of the physical mass for an unstable particle is rather non-trivial; however the decay rate Γ of the unstable particle state is usually very small compared to its lifetime, so that we can consider those as approximated eigenstates of the Hamiltonian.

We have to take into account the next facts:

- If $\Gamma \neq 0$ the pole of the two point function is not anymore on the real axis, but it is shifted in the complex plane, so that it acquires an imaginary part proportional to the decay rate Γ of the unstable particle state.
- If $\Gamma \ll m$, we may still define the physical mass m of the unstable particle state as the real part of the location of the pole of the two point function, since the imaginary part is

small compared to the real part, and it represents a small correction to the physical mass of the unstable particle state.

Thus in the end we can define the physical mass m of the unstable particle state as

$$m^2 = \mathrm{Re}\,[\text{propagator pole}]\,,$$

which gives us a well-defined physical mass for the unstable particle state, and we can write the two point function in momentum space as

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(0)\} |\Omega\rangle = \frac{iZ}{p^2 - m^2 + im\Gamma} + \Lambda(p^2),$$

where $\Lambda(p^2)$ is a regular function for $p^2 \sim m^2$ which encodes the regular contributions from the other states in the spectrum of the theory.

2.4.2 | LSZ Reduction Formula

TODO: Add reference to Schwartz 6.1

Let us start from the definition of the S-matrix elements in terms of the asymptotic in and out states:

$$\langle f| \hat{S} |i\rangle\,,$$

where $|i\rangle$ and $|f\rangle$ are the in and out states of the theory, which are defined as the asymptotic states of the theory at $t \to -\infty$ and $t \to +\infty$ respectively. Asymptotically, the interacting theory reduces to a free theory, so that the in and out states can be approximated by the free particle states of the free theory. This means that the field operators of the interacting theory can be approximated by the field operators of the free theory in this limit, in which they act as creation and annihilation operators for the free particle states (or 1-particle):

$$\hat{\phi}(x)\Big|_{t\to\pm\infty} \sim \sqrt{Z} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{p}}} \left(\hat{a}_\mathbf{p} e^{-ip\cdot x} + \hat{a}^\dagger_\mathbf{p} e^{ip\cdot x}\right),$$

where Z is the field strength renormalization constant of the one particle state in the spectrum of the theory, and $\hat{a}_\mathbf{p}$ and $\hat{a}^\dagger_\mathbf{p}$ are the annihilation and creation operators for the free particle states of the free theory. The energy appearing in the denominator is the physical energy of the one particle state, which is related to the physical mass m by the relativistic dispersion relation $E_\mathbf{p} = \sqrt{\mathbf{p}^2 + m^2} = p^0$.

Remark. *Note that the annihilation and creation operators $\hat{a}_\mathbf{p}$ and $\hat{a}^\dagger_\mathbf{p}$ are time independent in a free theory. We will assume that they are also time independent in the interacting theory, when considering the asymptotic limit $t \to \pm\infty$, even if in general*

$$\hat{a}_\mathbf{p}(t \to \infty) \neq \hat{a}_\mathbf{p}(t \to -\infty),$$

since the interactions can change the number of particles in the theory, so that the annihilation and creation operators at $t \to +\infty$ can be different from those at $t \to -\infty$ (the free-like field configurations may be different after the scattering process).

If we consider a scattering process with 2 incoming particles and n outgoing particles, we can write the in and out states as

$$|i\rangle = \sqrt{2E_a}\sqrt{2E_b}\hat{a}^\dagger_{\mathbf{p}_a}\hat{a}^\dagger_{\mathbf{p}_b} |\Omega\rangle\,,$$
$$|f\rangle = \sqrt{2E_1}\sqrt{2E_2}\cdots\sqrt{2E_n}\,\hat{a}^\dagger_{\mathbf{p}_1}\hat{a}^\dagger_{\mathbf{p}_2}\cdots\hat{a}^\dagger_{\mathbf{p}_n} |\Omega\rangle\,,$$

where we have properly normalized the states by the factors of $\sqrt{2E}$ according to the relativistic normalization convention. It is now time to compute the S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ by inserting the expressions for the in and out states and recalling that $\hat{S} = \mathbb{I} + i\hat{T}$, such that we can write

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle = \langle f|\,\mathbb{I} + i\hat{T}\,|i\rangle = \langle f|\,i\hat{T}\,|i\rangle = \sqrt{2E_a}\sqrt{2E_b}\sqrt{2E_1}\sqrt{2E_2}\cdots\sqrt{2E_n}\times \\ \times \langle\Omega|\,\hat{a}_{\mathbf{p_1}}(+\infty)\hat{a}_{\mathbf{p_2}}(+\infty)\cdots\hat{a}_{\mathbf{p}_n}(+\infty)\left(i\hat{T}\right)\hat{a}^\dagger_{\mathbf{p}_a}(-\infty)\,\hat{a}^\dagger_{\mathbf{p}_b}(-\infty)\,|\Omega\rangle\,.\end{aligned}$$

The idea is now to rewrite everyting in terms of Green's functions of the theory, so let us first prove the following identity:

$$\begin{aligned}&\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_x^2 + m^2)\hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\ldots,y_n)\} \\ &= \sqrt{2E_{\mathbf{p}}}\left(\hat{a}_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\ldots,y_n) - \hat{T}\hat{O}(y_1,\ldots,y_n)\hat{a}_{\mathbf{p}}(-\infty)\right),\end{aligned}$$

where $\hat{O}(y_1,\ldots,y_n)$ is an arbitrary product of operators of the theory, and we have used the fact that the field operator $\hat{\phi}(x)$ can be approximated by the free field operator in the asymptotic limit.

In order to prove this identity, we now define $\hat{\mathcal{O}}(x) = \hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\ldots,y_n)\}$ as the time ordered product of the field operator and the arbitrary product of operators, so that we can write

$$\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_x^2 + m^2)\hat{\mathcal{O}}(x) = \frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_t^2 - \partial_i^2 + m^2)\hat{\mathcal{O}}(x),$$

where we have explicitly written the derivatives in terms of the time and spatial derivatives. We can now integrate by parts the spatial derivatives, so that it changes sign and start acting on the left $\overrightarrow{\partial}_i \to \overleftarrow{\partial}_i$,[6] getting to

$$\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_t^2 + \mathbf{p}^2 + m^2)\hat{\mathcal{O}}(x),$$

where we can recognize the relativistic dispersion relation $E_{\mathbf{p}}^2 = \mathbf{p}^2 + m^2$ for a particle with mass m and momentum $\mathbf{p}$, so that we can write

$$\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, i\, e^{ip\cdot x}(\partial_t^2 + E_{\mathbf{p}}^2)\hat{\mathcal{O}}(x) = \frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, \partial_t\left(e^{ip\cdot x}(i\partial_t + E_{\mathbf{p}})\hat{\mathcal{O}}(x)\right).$$

Now we can split the integral in a spatial and a time integral, so that

$$\frac{1}{\sqrt{Z}}\int \mathrm{d}t\, \partial_t e^{iE_{\mathbf{p}}t}\int \mathrm{d}^3\mathbf{x}\, e^{-i\mathbf{p}\cdot\mathbf{x}}(i\partial_t + E_{\mathbf{p}})\hat{\mathcal{O}}(x),$$

which in this form is a total time derivative (the first integral), so that we can evaluate it at the boundaries $t = \pm\infty$, where we know the field operators to behave as free field operators, and the ladder operators to be time independent; so we can now work on the second integral and substitute the expression for $\hat{\mathcal{O}}(x)$ and the asymptotic behavior of the field operator, getting to

$$\int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\int \mathrm{d}^3\mathbf{x}\left[\frac{E_{\mathbf{p}} + E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}}e^{-iE_{\mathbf{k}}t}e^{-i(\mathbf{p}-\mathbf{k})\cdot\mathbf{x}}\hat{T}\{\hat{a}_{\mathbf{k}}\hat{O}\} + \frac{E_{\mathbf{p}} - E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}}e^{iE_{\mathbf{k}}t}e^{i(\mathbf{p}-\mathbf{k})\cdot\mathbf{x}}\hat{T}\{\hat{a}^\dagger_{\mathbf{k}}\hat{O}\}\right],$$

in which a spatial integration gives us a delta function $\delta^{(3)}(\mathbf{p}-\mathbf{k})$ which allows us to perform the integral over $\mathbf{k}$, getting to

$$\begin{aligned}\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, i\, e^{ip\cdot x}(\partial_t^2 + E_{\mathbf{p}}^2)\hat{\mathcal{O}}(x) &= \int \mathrm{d}t\, \partial_t\left[e^{iE_{\mathbf{p}}t}\sqrt{2E_{\mathbf{p}}}\left(e^{-iE_{\mathbf{p}}t}\hat{T}\{\hat{a}_{\mathbf{p}}(t)\hat{O}\}\right)\right] \\ &= \sqrt{2E_{\mathbf{p}}}\left[\hat{a}_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\ldots,y_n) - \hat{T}\hat{O}(y_1,\ldots,y_n)\hat{a}_{\mathbf{p}}(-\infty)\right],\end{aligned}$$

[6] We are assuming that the fields vanish at spatial infinity, so that the boundary terms from the integration by parts vanish: then the spatial derivatives will change sign and start acting on the left.

yielding the desired result. We can obtain a similar relation holding for its hermitian conjugate, which is

$$\begin{aligned}&\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\,(-i)\,e^{-ip\cdot x}(\partial_x^2+m^2)\hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\ldots,y_n)\}\\ &=\sqrt{2E_{\mathbf{p}}}\left[\hat{a}^\dagger_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\ldots,y_n)-\hat{T}\hat{O}(y_1,\ldots,y_n)\hat{a}^\dagger_{\mathbf{p}}(-\infty)\right].\end{aligned}$$

Now we can apply the derived result to the expression for the S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ that we wrote above, so that we can write

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle &= \sqrt{2E_a 2E_b 2E_1 2E_2\cdots 2E_n}\\ \times \langle\Omega|\,\hat{T}\hat{a}_{\mathbf{p}_1}(+\infty)\hat{a}_{\mathbf{p}_2}(+\infty)\cdots\hat{a}_{\mathbf{p}_n}(+\infty)&\left(i\hat{T}\right)\hat{a}^\dagger_{\mathbf{p}_a}(-\infty)\hat{a}^\dagger_{\mathbf{p}_b}(-\infty)\,|\Omega\rangle\,,\end{aligned}$$

where we inserted a time ordering operator T to use the derived result (the time ordering does not change the result since the ladder operators are already time-ordered). We can now apply the derived result for each of the ladder operators in the above expression: while we consider $\hat{a}_{\mathbf{p}_1}$ all the other ladder operators can be grouped in the arbitrary product of operators $\hat{O}$ in the derived result, so that we can apply it to each of the ladder operators in the above expression, getting to the famous **LSZ reduction formula**, which is essentially

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle &= \left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_a\,e^{ip_a\cdot x_a}(\partial^2+m^2)\phi(x_a)\right]\left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_b\,e^{ip_b\cdot x_b}(\partial^2+m^2)\phi(x_b)\right]\\ &\times\left[-\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_1\,e^{-ip_1\cdot x_1}(\partial^2+m^2)\phi(x_1)\right]\cdots\left[-\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_n\,e^{-ip_n\cdot x_n}(\partial^2+m^2)\phi(x_n)\right]\\ &\times\langle\Omega|\,T\{\phi(x)\phi(y)\ldots\phi(x_1)\phi(x_2)\ldots\phi(x_n)\}\,|\Omega\rangle\,,\end{aligned} \tag{2.4.6}$$

which is a way to relate the S-matrix elements to the time-ordered correlation functions of the fields, which are the Green's functions of the theory. The LSZ reduction formula is a powerful tool for computing the S-matrix elements from the Green's functions, and it is widely used in quantum field theory to compute scattering amplitudes and cross sections for various processes.

The LSZ reduction formula can be interpreted as a way to amputate the external legs of the Green's functions, which correspond to the free propagators of the theory, and to put them on shell by applying the operator (∂^2+m^2) to them, which gives zero when acting on the free propagators. The factors of $1/\sqrt{Z}$ in the formula are necessary to cancel the field strength renormalization constants that appear in the residues of the poles of the Green's functions, so that we can obtain finite results for the S-matrix elements.

Substantially, it is a bunch of Fourier transforms of the field operators, where the only difference is in the phase signs: incoming particles have a negative phase sign, while outgoing particles have a positive phase sign.

If we rewrite it in the momentum space, integrating by parts the derivatives so that they act on the exponentials on their left (which corresponds to the substitution $i(\partial_i^2+m^2)\to -i(p_i^2-m^2)$) we can write then the LSZ reduction formula as

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle &= \prod_{i=a,b,1,\ldots,n}\left[\frac{i}{\sqrt{Z}}(p_i^2-m^2)\right]\\ &\times\int \mathrm{d}^4x_a\,\mathrm{d}^4x_b\,\mathrm{d}^4x_1\,\cdots\,\mathrm{d}^4x_n\,e^{i(p_1\cdot x_1+\cdots+p_n\cdot x_n-p_a\cdot x_a-p_b\cdot x_b)}\\ &\times\langle\Omega|\,T\{\phi(x_1)\cdots\phi(x_n)\phi(x_a)\phi(x_b)\}\,|\Omega\rangle\,.\end{aligned} \tag{2.4.7}$$

We have assumed asymptotic particles as the final state, with on shell momenta $p_i^2 = m^2$, so that the factors of $(\partial^2 + m^2)$ in the LSZ reduction formula will give zero when acting on the external legs of the Green's functions, which are the free propagators of the theory. This means that the only contribution to the S-matrix elements will come from the poles of the Green's functions, which correspond to the physical particles in the spectrum of the theory:

$$\frac{iZ}{p_i^2 - m^2}.$$

The residues of these poles will give the field strength renormalization constants Z for each particle, which will cancel with the factors of $1/\sqrt{Z}$ in the LSZ reduction formula, giving a finite result for the S-matrix elements. Indeed, as $p_i \to \pm m$, the factors of $(p_i^2 - m^2)$ in the LSZ reduction formula will cancel with the poles of the Green's functions, giving a finite result for the S-matrix elements, since namely its just a series of regular terms for $p_i^2 \sim m^2$ multiplied by $\sqrt{Z}$.

2.4.3 | Generalization to Other Fields

If we repeat the same analysis for different types of fields, such as Dirac fields or gauge fields, we will find similar results for the two point functions and the LSZ reduction formula, with some modifications due to the different spin and statistics of the fields:

- For a **complex scalar field**, we would have two types of particles, the particle and the antiparticle, with different creation and annihilation operators, and the LSZ reduction formula would involve both types of operators for the in and out states:
  - $\phi^\dagger$ for *incoming particles* and *outgoing antiparticles*, with a positive phase sign in the Fourier transform.
  - ϕ for *outgoing particles* and *incoming antiparticles*, with a negative phase sign in the Fourier transform.

  This would lead to a more complicated LSZ reduction formula, but the general structure would be the same as for the real scalar field, and with this prescription we could also compute the results for Dirac theory ($\phi \to \psi$ and $\phi^\dagger \to \bar{\psi}$).

- If we are considering a **theory of several fields**, then the Green function will contain all the different fields along with the different masses of the quanta, and the LSZ reduction formula will involve the corresponding field operators for each type of particle in the in and out states, with the appropriate phase signs in the Fourier transforms.

- When considering **particle with different spins**, we have to take into account the different transformation properties of the fields under Lorentz transformations, which will affect the structure of the two point functions and the LSZ reduction formula. For example, for a Dirac field we would have the following field operators for the in and out states:
  - $\overline{u}_{s,\alpha}$ for *outgoing particles*;
  - $u_{s,\alpha}$ for *incoming particles*;
  - $v_{s,\alpha}$ for *outgoing antiparticles*;
  - $\overline{v}_{s,\alpha}$ for *incoming antiparticles.*

If we instead consider a spin-1 gauge field, we would have the following field operators for the in and out states:

- $\epsilon_\mu(p,\, \lambda)$ for *outgoing particles*;
- $\epsilon^*_\mu(p,\, \lambda)$ for *incoming particles.*

The spinorial/vectorial indices (α, μ etc.) of the field operators will be contracted with the corresponding indices of the Green's functions in the LSZ reduction formula, and the final expression for the S-matrix elements will involve the appropriate spinor/vector structures for each type of particle in the in and out states:

$$\langle\Omega|\, T\{\dots A^\mu(x)\dots\psi_\alpha(y)\dots\}\,|\Omega\rangle\,.$$

- For a Dirac fermion the factor $(\partial^2 - m^2)$ in the LSZ reduction formula will be replaced by the Dirac operator $(\not{\partial} - m)$, which gives zero when acting on the free propagator of the Dirac field, and the field strength renormalization constant Z will be replaced by the corresponding field strength renormalization constant for the Dirac field. Furthermore external factors $\not{p} - m$ will cancel out the poles of the Green's functions, leaving a factor of $\sqrt{Z}$ as $p_i \to$ *on-shell.*

We have to make one final observation: we did not make any assumption on the behaviour of the field operators of the interacting theory, apart from the fact that they can be approximated by the field operators of the free theory in the asymptotic limit (as ladder operat), which is roughly equivalent to the statement

$$\langle\Omega|\, \hat{\phi}(x)\,|\lambda_{\mathbf{p}}\rangle = \sqrt{Z}e^{-ip\cdot x},$$

whose validity was argued when we derived the Källén-Lehmann representation of the two point function. This means that the LSZ reduction formula is valid for any interacting theory, as long as the field operators of the interacting theory can be approximated by the field operators of the free theory in the asymptotic limit, which is a very general condition that is satisfied by most of the theories we are interested in.

Hence, our derivation is still valid when replacing the one particle state $|\mathbf{p}\rangle$ to any state $|\lambda_{\mathbf{p}}\rangle$ and the field operator $\hat{\phi}(x)$ to any operator $\hat{O}(x)$ of the theory, as long as the following condition is satisfied:

$$\langle\Omega|\, \hat{O}(x)\,|\lambda_{\mathbf{p}}\rangle \sim e^{-ip\cdot x},$$

which is valid as long as $\langle\Omega|\, \hat{O}(x)\,|\lambda_{\mathbf{p}}\rangle \neq 0$, so that the operator $\hat{O}(x)$ creates a one particle state from the vacuum with non-zero probability, and the LSZ reduction formula can be applied to any such operator. The LSZ reduction formula is a very general result that can be applied to a wide range of theories and operators, and it is a powerful tool for computing the S-matrix elements from the Green's functions of the theory.

Finally, inserting any operator $\hat{O}(x)$ among asymptotic states in the LSZ reduction formula is allowed in the form of the following substitutions:

$$\begin{cases}\langle f|\, \hat{S}\,|i\rangle = \langle f, t=+\infty|i, t=-\infty\rangle \\ \langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\}\,|\Omega\rangle\end{cases} \Longrightarrow \begin{cases}\langle f, t=+\infty|\, \hat{O}(x)\,|i, t=-\infty\rangle\,, \\ \langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\hat{O}(x)\}\,|\Omega\rangle\,,\end{cases}$$

respectively for the l.h.s and the r.h.s of the LSZ reduction formula. This means that we can compute matrix elements of any operator $\hat{O}(x)$ between asymptotic states by using the LSZ reduction formula, which is a very powerful result that allows us to compute a wide range of physical observables.

3 Feynman Diagrams

In the previous chapters we have seen how to calculate Green's functions in QFT and, as we have notice, the number of terms in the expansion of the Green's function grows very quickly as we increase the order of the correction, making it very difficult to keep track of all the terms and their contributions. This is where Feynman diagrams come in handy, as they provide a **visual representation of the interactions** and allow us to easily keep track of the different terms in the expansion.

In this chapter we will see how to derive a set of practical rules, known as Feynman rules, that allow us to compute Green's functions and scattering amplitudes in a systematic way for an *interacting theory in perturbation theory*, without having to calculate all the contractions explicitly. This method is based on Tylor expansions in the coupling constants g_j, assuming a weak enough interaction, and it is a very powerful tool for calculating physical quantities in QFT, as it allows us to write down the contribution of each diagram without having to calculate all the contractions explicitly, which is a very powerful tool for calculating physical quantities in QFT.

We will start by reviewing the Hamiltonian formalism and the interaction picture, which are essential for understanding how to calculate Green's functions in an interacting theory, and then we will see how to derive the Feynman rules for a simple interacting theory, and how to use them to calculate physical quantities. Through some examples, we will see how to apply the Feynman rules in practice, and then we will see how to generalize them to more complicated theories and to momentum space, which is the most common way to use Feynman diagrams in QFT.

3.1 | Hamiltonian Formalism and Interaction Picture

In QFT is often more convenient to work in the Hamiltonian formalism, where we express the dynamics of the system in terms of a Hamiltonian operator. The Hamiltonian can be split into a free part, which describes non-interacting particles, and an interaction part, which describes the interactions between particles.

The **Heisenberg picture** is often used in QFT, where operators evolve in time while states remain constant. However, for perturbative calculations, it is more convenient to use the **interaction picture**, where both operators and states evolve in time. Let us start from the Heisenberg equation of motion for an operator $\hat{O}(x)$:

$$i\partial_t \hat{O}(x) = [\hat{O}(x), \hat{H}(x)],$$

where $\hat{H}(x)$ is the Hamiltonian operator. We can write the solution to this equation through the time evolution operator $\hat{S}(t, t_0)$ as

$$\hat{O}(\mathbf{x}, t) = \hat{S}^\dagger(t, t_0)\hat{O}(\mathbf{x}, t_0)\hat{S}(t, t_0),$$

which can be inserted back into the Heisenberg equation of motion to get an equation for the time evolution operator:

$$i\partial_t \hat{S}(t, t_0) = \hat{H}(t)\hat{S}(t, t_0).$$

Now let us split the Hamiltonian into a free part and an interaction part:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t),$$

and the Lagrangian into a free part and an interaction part:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}.$$

In practice, $\hat{H}_0$ will be the Hamiltonian of a free theory, which describes non-interacting particles whose dynamics can be solved exactly, while $\hat{V}(t)$ will be the interaction Hamiltonian, which will be treated as a perturbation and encodes all the interaction terms. The free part of the Lagrangian $\mathcal{L}_0$ will describe the dynamics of the free fields, while the interaction part $\mathcal{L}_{\text{int}}$ will describe the interactions between the fields, indeed we can write the Legendre transform of the Lagrangian to get the Hamiltonian as

$$H = \int \mathrm{d}^3\mathbf{x}\, \mathcal{H}(\mathbf{x}), \qquad \begin{cases} \mathcal{H} = \pi\dot{\phi} - \mathcal{L}, \\ \pi = \partial\mathcal{L}/\partial\dot{\phi}, \end{cases}$$

and if $\mathcal{L}_{\text{int}}$ does not contain derivatives, then we simply have $\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}}$, and thus we can write the interaction Hamiltonian as

$$V = -\int \mathrm{d}^3\mathbf{x}\, \mathcal{L}_{\text{int}}.$$

If $\mathcal{L}_{\text{int}}$ does indeed contain derivatives, then similar results can be obtained, but it will be easier to derive them in the Lagrangian formalism, which is more convenient for theories with derivative interactions, and we will see how to do that in the next chapters.

3.1.1 | Interaction Picture

We are now moving from the Heisenberg picture to the interaction picture, where both operators and states evolve in time. In the interaction picture, the *fields evolve according to the free Hamiltonian H_0*, while the *states evolve according to the total Hamiltonian V*: the interaction is encoded in the evolution of the states, while the fields evolve as if they were free fields. This is a very important point, as it allows us to use the free fields to calculate the interactions, which is much easier than dealing with the full interacting fields.

Then of course the *matrix elements* we are interested in computing *will be independent from the chosen picture*, since they are physical quantities, and thus we can choose the picture that is more convenient for our calculations. Schrödinger, Heisenberg and interaction picture are just different ways of describing the same physics, and they are identically equivalent equivalent at a conventional reference time $t = t_0$.

Thus we can start from the Schrödinger picture, where the fields are time-independent, and then we can move to the interaction and Heisenberg picture, where the fields evolve in time according to the free Hamiltonian and the total Hamiltonian respectively. We can write the fields in the different pictures and see how they are related to each other as

$$\begin{cases} \hat{\phi}_0 = \hat{\phi}_I = e^{i\hat{H}_0(t-t_0)}\hat{\phi}_S e^{-i\hat{H}_0(t-t_0)}, \quad \hat{\phi}_S = \hat{\phi}_I(t = t_0) \\ \hat{\phi} = \hat{\phi}_H = \hat{S}^\dagger(t, t_0)\hat{\phi}_S\hat{S}(t, t_0) = \hat{U}^\dagger(t, t_0)\hat{\phi}_I\hat{U}(t, t_0), \end{cases}$$

where

$$\hat{U}(t, t_0) = e^{i\hat{H}_0(t-t_0)}\hat{S}(t, t_0),$$

is the time evolution operator connecting the interaction picture to the Heisenberg picture, while $\hat{S}(t, t_0)$ is the time evolution operator connecting the Schrödinger picture to the Heisenberg picture.

Now having $\hat{H}_H = \hat{H}_S$ and $\hat{H}_0 = \hat{H}_{0,S}$ (i.e. the Heisenberg Hamiltonian is the same as the Schrödinger Hamiltonian, and the free Hamiltonian is the same in both pictures) we can write the equation of motion for $\hat{U}$ by inserting the fields in the Heisenberg equation of motion, which gives

$$i\partial_t\hat{U}(t, t_0) = \hat{V}_I(t, t_0)\hat{U}(t, t_0),$$

with

$$\hat{V}_I = e^{i\hat{H}_0(t-t_0)}\hat{V}_S e^{-i\hat{H}_0(t-t_0)},$$

which is the **interaction potential** (potential in the interaction picture). In practice, since $\hat{V}$ is given by the integral of the interaction Lagrangian, we can write it as

$$\hat{V}_I = \hat{V}_S(\hat{\phi})\big|_{\hat{\phi}\to\hat{\phi}_I} = \hat{V}(\hat{\phi}_0),$$

which means that the potential in the interaction picture is the same as the potential in the Schrödinger picture, but with the fields replaced by their interaction picture counterparts (which is the same as the free fields). This is a very important result, as it allows us to use the free fields to calculate the interactions, which is much easier than dealing with the full interacting fields.

Dyson Series

The solution to the evolution equation for $\hat{U}$ can be written as the following well-known time-ordered exponential:

$$\hat{U}(t, t_0) = \hat{T}\exp\left(-i\int_{t_0}^{t} \mathrm{d}t'\hat{V}_I(t')\right),$$

which can be expanded in a power series as

$$\hat{U}(t, t_0) = \mathbb{I} + (-i)\int_{t_0}^{t} \mathrm{d}t'\hat{V}_I(t') + \frac{(-i)^2}{2!}\int_{t_0}^{t} \mathrm{d}t'\int_{t_0}^{t} \mathrm{d}t''\hat{T}\{\hat{V}_I(t')\hat{V}_I(t'')\} + \dots$$

where if we now partition the integration domain of the n-th term of the series in a set of $n!$ subdomains, with the idea to apply the time-ordering operator: up to renaming the integration variables, we can reorder the integration limits as

$$t' > t'' > \cdots > t^{(n)} > t_0,$$

making the integral of the n-th term of the series as

$$\hat{U}(t,t_0) = \mathbb{I} + (-i)\int_{t_0}^{t} \mathrm{d}t' \hat{V}_I(t') + (-i)^2 \int_{t_0}^{t} \mathrm{d}t' \int_{t_0}^{t'} \mathrm{d}t'' \hat{V}_I(t')\hat{V}_I(t'') + \ldots$$

which is known as the **Dyson series**, and it is the solution to the evolution equation for $\hat{U}$. Thus we can write the Dyson series in such a way that the time-ordering operator is not needed, since the integration limits ensure that the operators are ordered correctly in time. In the end we have

$$\hat{U}(t,t_0) = \mathbb{I} - i\int_{t_0}^{t} \mathrm{d}t' \hat{V}_I(t')\hat{U}(t',t_0),$$

which is an integral equation for $\hat{U}$ that can be solved iteratively, and we can see that the Dyson series is the solution to this integral equation. This is a very important result, as it allows us to calculate the time evolution operator in the interaction picture through

$$i\partial_t \hat{U}(t,t_0) = \hat{V}_I(t)\hat{U}(t,t_0),$$

which is essential for calculating scattering amplitudes and other physical quantities in QFT, even if the computation remains quite complicated.

3.1.2 | Computing Green's Functions

Consider a Green's function of the form

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\, |\Omega\rangle\,,$$

where $|\Omega\rangle$ is the vacuum state of the interacting theory, and $\hat{\phi}(x) = \hat{\phi}_H(x)$ are the interacting fields in the Heisenberg picture. Since the matrix elements are independent from the chosen picture, we can write it in the interaction picture as

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\, |\Omega\rangle = \frac{\langle 0|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\exp\Big[-i\int_{-\infty}^{\infty} \mathrm{d}t\, \hat{V}_I(t)\Big]\}\, |0\rangle}{\langle 0|\, \hat{T}\{\exp\Big[-i\int_{-\infty}^{\infty} \mathrm{d}t\, \hat{V}_I(t)\Big]\}\, |0\rangle},$$

where we have used the fact that the vacuum state in the interaction picture is the same as the vacuum state in the Schrödinger picture, and we have also used the fact that the time evolution operator in the interaction picture can be written as a time-ordered exponential of the interaction Hamiltonian.

We can expand the computation considering that the interaction Hamiltonian is given by the integral of the interaction Lagrangian, which is the case for a large class of theories, and thus we can write

$$\mathcal{H}_{int} = -\mathcal{L}_{int}, \qquad \int \mathrm{d}t\, V_I(t) = -\int \mathrm{d}^4x\, \mathcal{L}_{int}(x),$$

so that the Green's function can be written as

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\, |\Omega\rangle = \frac{\langle 0|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\exp\Big[i\int \mathrm{d}^4x\, \hat{\mathcal{L}}_{int}\Big]\, |0\rangle}{\langle 0|\, \hat{T}\{\exp\Big[i\int \mathrm{d}^4x\, \hat{\mathcal{L}}_{int}\Big]\}\, |0\rangle},$$

which is a very important result, as it allows us to calculate Green's functions in the interaction picture, which is much easier than calculating them in the Heisenberg picture, since we can use the free fields to calculate the interactions. This is the basis for perturbation theory in QFT, where we can expand the exponential of the interaction Lagrangian and calculate the Green's functions order by order in perturbation theory.

3.1.3 | Wick's Theorem

Wick's theorem is a fundamental result in quantum field theory that allows us to express the time-ordered product of field operators as a sum of normal-ordered products and contractions. It is particularly useful for calculating Green's functions and scattering amplitudes in perturbation theory. Let us consider a time-ordered Green's function of the form

$$\langle 0| \hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\} |0\rangle ,$$

where $\hat{\phi}(x)$ is a free scalar field operator, thus behaving as a ladder operator. We can manipulate the fields as

$$\phi = \phi^+ + \phi^-, \qquad \begin{cases} \phi^+ |0\rangle = 0, \\ \langle 0| \phi^- = 0, \end{cases}, \qquad \begin{cases} \phi^+ = \displaystyle\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_\mathbf{p}}} a_\mathbf{p} e^{-ip\cdot x}, \\ \phi^- = \displaystyle\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_\mathbf{p}}} a^\dagger_\mathbf{p} e^{ip\cdot x}, \end{cases}$$

where ϕ^+ is the annihilation part of the field, and ϕ^- is the creation part of the field. We can then use the normal ordering of the product of $\hat{a}$ and $\hat{a}^\dagger$ to simplify the expression: it puts all the creation operators to the left of the annihilation operators, thus ensuring that the vacuum expectation value of any normal-ordered product is zero. We will use the notation $:\hat{O}:$ to denote the normal ordering of an operator (or a general product of operators) $\hat{O}$.

Example. Let us consider the product of four ladder operators:

$$a(p_1)a^\dagger(p_2)a(p_3)a^\dagger(p_4),$$

then we can normal order this product as

$$:a(p_1)a^\dagger(p_2)a(p_3)a^\dagger(p_4):= a^\dagger(p_2)a^\dagger(p_4)a(p_1)a(p_3),$$

and we can see that the vacuum expectation value of this normal-ordered product is zero:

$$\langle 0| :a(p_1)a^\dagger(p_2)a(p_3)a^\dagger(p_4): |0\rangle = 0.$$

If we now consider the time-ordered product of two fields, a **2-point T-product**, assuming $x^0 > y^0$, we can write it as

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}_{x^0>y^0} = \hat{\phi}^+(x)\hat{\phi}^+(y) + \hat{\phi}^+(x)\hat{\phi}^-(y) + \hat{\phi}^-(x)\hat{\phi}^+(y) + \hat{\phi}^-(x)\hat{\phi}^-(y),$$

where the only term that is not normal-ordered is the second term. We can then write the normal-ordered product of the two fields as

$$:\hat{\phi}(x)\hat{\phi}(y):= \hat{\phi}^-(x)\hat{\phi}^-(y) + \hat{\phi}^-(x)\hat{\phi}^+(y) + \hat{\phi}^-(y)\hat{\phi}^+(x) + \hat{\phi}^+(x)\hat{\phi}^+(y),$$

which enforces

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}_{x^0>y^0} =: \hat{\phi}(x)\hat{\phi}(y): + [\hat{\phi}^+(x), \hat{\phi}^-(y)].$$

Since a symmetric expression exists for the case $x^0 < y^0$, we can write in general that the time-ordered product of two fields can be written as:

$$\hat{t}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) : +\theta(x^0 - y^0)[\hat{\phi}^+(x), \hat{\phi}^-(y)] + \theta(y^0 - x^0)[\hat{\phi}^+(y), \hat{\phi}^-(x)].$$

If we now take the vacuum expectation value of this time-ordered product (the commutator is not an operator, it is just a function multiplies by the identity), we can see that the normal-ordered part will vanish, since it contains only properly ordered creation and annihilation operators, and thus we are left with

$$\langle 0| \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} |0\rangle = \Theta(x^0 - y^0)[\hat{\phi}^+(x), \hat{\phi}^-(y)] + \Theta(y^0 - x^0)[\hat{\phi}^+(y), \hat{\phi}^-(x)] = D_F(x, y),$$

where $D_F(x, y)$ is the Feynman propagator, describing the propagation of particles between two points in spacetime:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) : +D_F(x, y).$$

Thus if we define the **contraction of two fields** as *the vacuum expectation value of the time-ordered product* of those two fields, which is the Feynman propagator

$$\underbrace{\hat{\phi}_1\hat{\phi}_2} = \langle 0| \hat{T}\{\hat{\phi}_1\hat{\phi}_2\} |0\rangle = D_F(x_1, x_2),$$

we can write the time-ordered product of two fields as the normal-ordered product of those two fields summed to the contraction between them:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) + \underbrace{\hat{\phi}(x)\hat{\phi}}(y) :$$

where the contraction is denoted by a line connecting the two fields, and it represents the Feynman propagator between those two fields. This is the simplest form of Wick's theorem, which allows us to express the time-ordered product of two fields as a sum of a normal-ordered product and a contraction.

This was the Wick's theorem for the time-ordered product of two fields, but we can generalize for it induction to the time-ordered product of n fields, which is the most general form of Wick's theorem:

Theorem 3.1 (Wick's theorem)**.** *The time-ordered product of n fields can be written as the normal-ordered product of those n fields plus all possible contractions between those fields:*

$$\hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\} = : \hat{\phi}(x_1)\ldots\hat{\phi}(x_n) + \textit{ all possible contractions } :$$

where a contraction between two fields is defined as the vacuum expectation value of the time-ordered product of those two fields, which is the Feynman propagator between those two fields:

$$\underbrace{\hat{\phi}_1\hat{\phi}_2} = \langle 0| \hat{T}\{\hat{\phi}_1\hat{\phi}_2\} |0\rangle = D_F(x_1, x_2).$$

Example (4-point function). Let us consider the following 4-point function

$$\langle 0| \hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\} |0\rangle ,$$

and let us apply Wick's theorem to it. We can write the time-ordered product of the four fields as the normal-ordered product of those four fields plus all possible contractions between those fields:

$$\begin{aligned}\hat{T}\{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4\} = &: \hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 + \\ &+ \underbrace{\hat{\phi}_1\hat{\phi}_2}\hat{\phi}_3\hat{\phi}_4 + \underbrace{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3}\hat{\phi}_4 + \ldots \\ &+ \hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 + \underbrace{\hat{\phi}_1\hat{\phi}_2}\underbrace{\hat{\phi}_3\hat{\phi}_4} + \underbrace{\hat{\phi}_1\underbrace{\hat{\phi}_2\hat{\phi}_3}\hat{\phi}_4} :\end{aligned}$$

where we denoted $\hat{\phi}(x_i) \equiv \hat{\phi}_i$. Note that when contracting two fields in a product of more than two fields, we do not need to worry about having adjacent fields, since the contraction is well defined as

$$: \hat{\phi}_1 \hat{\phi}_2 \hat{\phi}_3 \hat{\phi}_4 : = : \hat{\phi}_2 \hat{\phi}_4 : D_F(x_1, x_3),$$

and thus we can contract any two fields in the product, regardless of their position, and we will always get a well-defined result.

Using this last observation, we can notice how all the contractions of only two fields will give us a normal-ordered product of the remaining two fields, which will have zero vacuum expectation value, thus the only terms that will contribute to the vacuum expectation value of the 4-point function are the terms with two contractions, which will give us a product of two Feynman propagators: taking all the possible contractions of two pairs of fields, we get

$$\begin{aligned} \langle 0| \hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\} |0\rangle &= D_F(x_1, x_2) D_F(x_3, x_4) \\ &+ D_F(x_1, x_3) D_F(x_2, x_4) + D_F(x_1, x_4) D_F(x_2, x_3), \end{aligned}$$

which shows the extent to which the number of terms grows as we increase the number of fields, since we have to sum over all possible contractions.

Wick's theorem can be easily proven by induction, starting from the case of two fields, which we have already shown, and then assuming that it holds for $n-1$ fields, we can show that it holds for n fields by writing the time-ordered product of n fields as the time-ordered product of the first field and the time-ordered product of the remaining $n-1$ fields, and then applying the induction hypothesis to the second part. Let us now see some important consequences of Wick's theorem.

Corollary 3.2. *As we said before, the vacuum expectation value of any normal-ordered product is zero, thus we have*

$$\langle 0| : \hat{\phi}(x_1) \dots \hat{\phi}(x_n) : |0\rangle = 0,$$

and thus we can write the vacuum expectation value of the time-ordered product of n fields as

$$\langle 0| \hat{T}\{\hat{\phi}(x_1) \dots \hat{\phi}(x_n)\} |0\rangle = \textit{all possible contractions},$$

where all possible contractions means that we need to sum over all possible ways of contracting the fields, and each contraction gives us a Feynman propagator. This is a very important result, as it allows us to calculate Green's functions in terms of Feynman propagators, which are much easier to calculate than the full time-ordered product of fields.

Corollary 3.3. *If in a free theory we have an odd number of fields, then there are no possible contractions, since we cannot contract an odd number of fields, thus we have*

$$\langle 0| \hat{T}\{\hat{\phi}(x_1) \dots \hat{\phi}(x_{2n+1})\} |0\rangle = 0,$$

which is a very important result, as it tells us that the vacuum expectation value of the time-ordered product of an odd number of fields is always zero, which is a consequence of the fact that we cannot contract an odd number of fields.

This last result can be proven using the fact that the normal-ordered product of an odd number of fields is zero, since it contains an odd number of creation and annihilation operators, and thus its vacuum expectation value is zero, and thus we are left with only the contractions, which are also zero since we cannot contract an odd number of fields. For an interacting theory, this is not necessarily true, since the interaction can create particles and thus we can have non-zero vacuum expectation values for the time-ordered product of an odd number of fields, but in a free theory this is always zero.

3.2 | Feynman Diagrams and Rules

Let us start to put all the pieces together with an example. We will first derive the Feynman rules for a simple interacting theory in the coordinate space, and then we will see how to generalize them to more complicated theories and to momentum space.

3.2.1 | Coordinates Space

Example (ϕ^3 theory). Let us study a ϕ^3 theory, which is a simple interacting theory of a scalar field with a cubic interaction term in the Lagrangian:

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{3!}\phi^3,$$

where λ is the coupling constant that controls the strength of the interaction. We want to calculate the 2-point function in this theory, which is given by

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\}\,|\Omega\rangle\,,$$

where $|\Omega\rangle$ is the vacuum state of the interacting theory and $\hat{\phi}$ are the field operators. Using the formula we derived before for the Green's function in the interaction picture, we can write this as

$$\frac{\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\}\exp\left[i\int \mathrm{d}^4x\,\mathcal{L}_{\text{int}}\right]|0\rangle}{\langle 0|\,\hat{T}\{\exp\left[i\int \mathrm{d}^4x\,\mathcal{L}_{\text{int}}\right]\}\,|0\rangle},$$

with the free field operators $\hat{\phi}_0$; now the idea is to expand the exponential in a power series. Let us start from the numerator, which we can write as

$$\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\}\,|0\rangle + (-i\frac{\lambda}{3!})\,\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\}\int \mathrm{d}^4x\,\hat{\phi}_0(x)\,|0\rangle +$$
$$+(-i\frac{\lambda}{3!})^2\,\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\}\int \mathrm{d}^4x\,\hat{\phi}_0(x)\int \mathrm{d}^4y\,\hat{\phi}_0(y)\,|0\rangle + \dots$$

where we have expanded the exponential of the interaction Lagrangian in a power series, and we have also used the fact that the fields in the interaction picture are the same as the free fields, which we have denoted with a subscript 0. With $\lambda = 0$ we get the free theory back, with the usual Feynman propagator as we expected.

The first term in the expansion thus corresponds to the free theory, which is just a single line connecting the two external points x_1 and x_2, representing the propagation of a particle from x_1 to x_2. The second term corresponds to a correction to the free theory, which can be represented as a diagram with a loop, where the particle propagates from x_1 to x, interacts at x with another particle, and then propagates from x to x_2. The third term corresponds to a correction with two loops, where the particle propagates from x_1 to x, interacts at x with another particle, then propagates from x to y, interacts at y with another particle, and then propagates from y to x_2. And so on for higher-order corrections.

If λ remains small, we can truncate the expansion at a certain order and get an approximation for the 2-point function. This is the basis for perturbation theory in QFT, where we can calculate physical quantities as a power series in the coupling constant, and each term in the series corresponds to a Feynman diagram that represents a particular process or interaction.

Let us highlight the fact that, for the Wick's theorem, the correction of the order $O(\lambda)$ vanishes, since it contains an odd number of fields, and thus there are no possible contractions, and thus its vacuum expectation value is zero. The first non-vanishing correction to the free theory is thus the second order correction, which is of order $O(\lambda^2)$, and it corresponds to a diagram with a loop, as we have seen before.

Let us introduce a simpler **shortcut notation** for the following computations, since we will have to deal with a large number of fields and propagators, and it will be easier to keep track of them in this way. We will denote the Feynman propagator between two external points x_i and x_j as D_{ij}, the Feynman propagator between an external point x_i and an internal point x as D_{ix}, and the free field operator at a point x_i as ϕ_i. Thus we have

$$\begin{cases} D_{ij} = D_F(x_i, x_j), \\ D_{xi} = D_F(x, x_i), \\ \phi_i = \phi_0(x_i), \end{cases}$$

so that we can write the second order correction to the free theory as

$$(-\frac{i\lambda}{3!})^2 \frac{1}{2!} \int \mathrm{d}^4 z_1 \, \mathrm{d}^4 z_2 \, \langle 0 | \hat{T}\{\hat{\phi}_x \hat{\phi}_y \hat{\phi}_{z_1} \hat{\phi}_{z_1} \hat{\phi}_{z_1} \hat{\phi}_{z_2} \hat{\phi}_{z_2} \hat{\phi}_{z_2}\} | 0 \rangle \,,$$

where now we can use Wick's theorem to calculate the vacuum expectation value of this time-ordered product, which will give us a sum of all possible contractions between the fields:

$$(-\frac{i\lambda}{3!})^2 \frac{1}{2!} \int \mathrm{d}^4 x \, \mathrm{d}^4 y \, (D_{1x} D_{2y} D_{xy}^2 + \dots),$$

where we have to note that the number of contractions grows very quickly and different contractions may give us the same contribution, since they may be equivalent under permutations of the fields.

But maybe there is a way in which we can **list all the possible contractions** without having to write them all down, since as we can see, the number of contractions grows very quickly as we increase the order of the correction, and they may also be equivalent. This is where Feynman diagrams come in handy, as they provide a visual representation of the interactions and allow us to easily keep track of the different terms in the expansion.

Now we can start to derive the Feynman rules for this theory, which will allow us to write down the contribution of each diagram without having to calculate all the contractions explicitly. The Feynman graphical representation can be derived from this example, where we have the following elements:

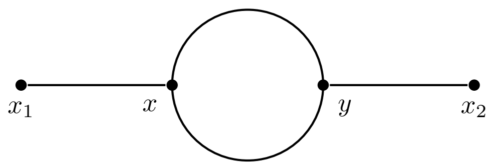


Figure 3.1: Feynman diagram for the second order correction to the 2-point function in a ϕ^3 theory: each internal point corresponds to an interaction vertex, connected to three lines, each corresponding to a propagator.

- x_1, x_2 external points, represented as dots.
- x, y internal points, represented as vertices.
- D_{x1} and D_{y2} propagators, represented as lines connecting the external points to the internal points.
- D_{xy}^2 propagators, represented as two lines connecting the internal points x and y.

More generally, we can write the **Feynman rules** for an n-point function as

- N external points $x_1, x_2, \ldots, x_n$, each connected to at least a propagator.

- Any number of vertexes, each corresponding to an interaction term in the Lagrangian, and thus
$$i \int \mathrm{d}^4 z \, \mathcal{L}_{\text{int}}(z),$$
where in particular for our ϕ^3 theory, we have
$$i \int \mathrm{d}^4 z \, (-\frac{\lambda}{3!}\phi^3(z)) = -i\frac{\lambda}{3!} \int \mathrm{d}^4 z \, \phi^3(z),$$
which means that each vertex has to be connected to three lines, since the interaction term is cubic in the field. Every vertex is of order $O(\lambda)$, since it comes from the interaction Lagrangian, and thus the order of the correction is given by the number of vertexes in the diagram.

- The propagators as lines connecting ALL the points, since the contractions has to be complete, and thus we need to connect all the points with propagators, since we cannot have uncontracted fields in the vacuum expectation value.

These rules telling us how to draw the Feynman diagrams for a given process, and they also allow us to write down the mathematical expression corresponding to each diagram without having to calculate all the contractions explicitly, which is a very powerful tool for calculating physical quantities in QFT.

The procedure of calculating a physical quantity using Feynman diagrams can be summarized as follows:

1. Draw all the possible Feynman diagrams for the process you want to calculate, following the Feynman rules.

2. For each vertex, write down the corresponding factor from the interaction Lagrangian $\rightarrow$ $-i\alpha \int \mathrm{d}^4 z$.

3. For each propagator, write down the corresponding Feynman propagator $D_F(x, y)$.

Diagram Prefactors

We have to understand how to put the correct prefactors α in front of the integrals. Let us start from the interaction Lagrangian, which is given by
$$\mathcal{L}_{\text{int}} = \frac{\lambda^N}{N!}\phi^N,$$
then a diagram with M vertexes (obtained considering corrections up to order $O(\lambda^M)$) and any number of external points will have

- $M!$ ways of permuting the vertexes, since they are indistinguishable.

- $N!$ ways of permuting the fields in each vertex, since they are also indistinguishable, thus a total of $(N!)^M$ ways of permuting the fields in all the vertexes.

Putting all these factors together, we get a factor of $(N!)^M \times M!$ only from the counting of the permutations of the vertexes and fields, and thus we need to divide by this factor to avoid overcounting, since we are summing over all the possible contractions, which already takes into account all the possible permutations of the vertexes and fields.

An example of naive counting of the factors for a diagram with M vertexes and N fields in each vertex would give us a factor of

$$(N!)^M \times M! \times (\frac{\lambda}{N!})^M \times \frac{1}{M!},$$

where the first factor comes from the permutations of fields in each vertex, the second factor comes from the permutations of the vertexes, the third factor comes from the interaction Lagrangian, and the last factor comes from the expansion of the exponential of the interaction Lagrangian. Is it possible that all these factors cancel out and we are left with a factor of λ^M for a diagram with M vertexes?

No, this counting is not correct, since we have to take into account that some diagrams may be equivalent under permutations of the vertexes and fields, and thus we need to divide by the number of symmetries of the diagram: *the number of permutations of the vertexes and fields that leave the diagram unchanged.* This is known as the **symmetry factor** of the diagram, and it can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram.

Thus the correct factor for a diagram with M vertexes and N fields in each vertex is given by the inverse of the symmetry factor of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram:

Definition 3.1 (Symmetry factor)**.** The symmetry factor of a Feynman diagram is the **number of independent transformations** (permutations of the vertexes and fields in the diagram) **mapping the diagram onto itself** (with external points fixed), and it is denoted by S. The contribution of a diagram to a physical quantity is given by the inverse of the symmetry factor, which is $1/S$. The identity has to be counted as a symmetry, while swapping external points is not a symmetry, since they are fixed. One can sum the symmetries of the drawing by hand, or one can multiply the symmetry prefactors of the subdiagrams (D_{xy}, D^2_{xx}...).

Example (Back to the cubic interaction). Let us go back to the second order correction to the 2-point function in the ϕ^3 theory, which corresponds to the following diagram:

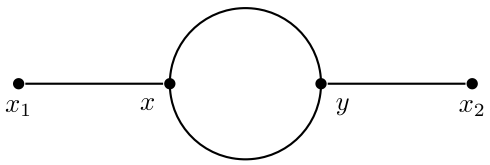


Figure 3.2: Feynman diagram for the second order correction to the 2-point function in a ϕ^3 theory: each internal point corresponds to an interaction vertex, connected to three lines, each corresponding to a propagator.

This diagram has two vertexes, each connected to three lines, and thus it corresponds to a correction of order $O(\lambda^2)$. It can be obtained from 18 different contractions:

- 3 choices of ϕ_x in $\underbrace{\phi_1\phi_x}$;
- 3 choices of ϕ_y in $\underbrace{\phi_2\phi_y}$;
- 2 choices of the remaining ϕ_y to be contracted with the first remaining ϕ_x.

These 18 contractions have to be multiplied by 2 for the swap of the two vertexes $x \leftrightarrow y$. Thus we get a total of 36 contractions which has to be divided by $(3!)^2 \times 2! = 72$, which is the number of permutations of the fields in the vertexes and the number of permutations of the vertexes, thus we get a symmetry factor of $S = 2$ for this diagram, and thus its contribution to the 2-point function is given by

$$\frac{1}{2} \times (-i\frac{\lambda}{3!})^2 \int \mathrm{d}^4 z_1 \, \mathrm{d}^4 z_2 \, D_{xz_1} D_{yz_1} D^2_{z_1 z_2},$$

where the factor of $1/2$ comes from the symmetry factor of the diagram, and the rest of the factors come from the interaction Lagrangian and the Feynman propagators.

In order to understand the symmetry factor of this diagram, we can also look at it from a different perspective; if we rewrite the contraction, we can see that permuting the propagators between the internal points x and y does not change the diagram, since they are indistinguishable:

$$\phi_1\phi_2\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y \quad = \quad \phi_1\phi_2\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y,$$

and if we compare it to the graphical representation of the diagram, we can see that this contraction symmetry corresponds to the symmetry of the diagram under the **vertical flip symmetry**, which is a symmetry of the diagram that leaves the external points fixed and swaps the internal points x and y. This symmetry is the only non-trivial symmetry of the diagram and thus we have a total of 2 symmetries (do not forget the identity), which gives us a symmetry factor of $S = 2$, and thus a contribution of $1/2$ for this diagram.

It can even happen to have an internal subdiagram that is disconnected from the external points, which is called a *vacuum bubble*, and it is possible to show that the contribution of a vacuum bubble cancels out with the denominator of the Green's function. For example, if we compute the next order corrections to the 2-point function, we get [... diagrams ...] where the fifth and sixth diagrams have vacuum bubble. The complete expression associated to these diagrams is given by

$$\begin{aligned}\langle\Omega|\, \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\, |\Omega\rangle = & D_{xy} + (-i\frac{\lambda}{3!})^2 \int \mathrm{d}^4 z_1 \, \mathrm{d}^4 z_2 \, D_{xz_1} D_{yz_1} D^2_{z_1 z_2} + \\ & + \dots\end{aligned}$$

where the contribution of the vacuum bubbles has been canceled out with the denominator of the Green's function, which is given by

$$\frac{1}{\langle 0|\, \hat{T}\{\exp\left[i \int \mathrm{d}^4 z \, \mathcal{L}_{int}\right]\}\, |0\rangle}.$$

Thus the denominator is exactly a Green's function with no external points, and thus it contains only **vacuum diagrams**, which are disconnected from the external points, and thus they do not contribute to the physical quantity we are calculating, and thus they can be canceled out with the denominator. This is a very important result, as it tells us that we can ignore the vacuum bubbles when calculating physical quantities in QFT, since they do not contribute to the final result, and thus we can focus only on the connected diagrams, which are the ones that contribute to the physical quantity we are calculating. Indeed we have:

- Numerator: ($\sum$ connected diagrams) ($1 + \sum$ vacuum diagrams).
- Denominator: ($1 + \sum$ vacuum diagrams).

Thus in the end, the contribution of the vacuum diagrams cancels out, and we are left with only the connected diagrams, which are the ones that contribute to the physical quantity we are calculating:

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\cdots\hat{\phi}(z)\}\, |\Omega\rangle = \sum \text{connected diagrams}.$$

Summary of Feynman Rules in Position Space

Let us group and summarize all the Feynman rules we have derived so far for a scalar field theory in position space:

TODO: Add more details.

- External point: x_i, represented as a dot, and connected only to one propagator.
- Vertex: $i \int \mathrm{d}^4 z\, \mathcal{L}_{\text{int}}(z)$, represented as a dot and connected to a number of lines equal to the number of fields in the interaction term.
- Propagator: $D_F(x, y)$, represented as a line connecting two points, and corresponding to the Feynman free propagator between those two points.
- Symmetry factor: $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram.
- $\langle\Omega| \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\cdots\hat{\phi}(z)\} |\Omega\rangle = \sum$ connected diagrams, since the contribution of the vacuum diagrams cancels out with the denominator of the Green's function, and thus we can focus only on the connected diagrams, which are the ones that contribute to the physical quantity we are calculating.

3.2.2 | Momentum Space

Let us study the same ϕ^3 theory $\mathcal{L}_{int} = \frac{\lambda}{3!}$ for a $2 \to 2$ scattering process. We work up to the **leading order** (LO) correction, which means up to $O(\lambda)$: only one vertex, the first non zero matrix element. The Green's function we are considering is given by

TODO: add draw of scatter.

$$\begin{aligned}
\langle f| \hat{S} |i\rangle &= \langle p_3, p_4| \hat{S} |p_1, p_2\rangle \\
&= \prod_{j=1}^{4} \left(-\frac{i}{\sqrt{Z}} (p_j^2 - m^2) \right) \\
&\times \int \mathrm{d}^4 x_1 \,\dots\, \mathrm{d}^4 x_4 e^{i(p_3 x_3 + p_4 x_4 - p_1 x_1 - p_2 x_2)} \\
&\times \langle\Omega| \hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\} |\Omega\rangle\,,
\end{aligned}$$

where we have used the LSZ reduction formula to relate the S-matrix element to the Green's function. Now for a free theory $Z = 1$, while for an interacting theory $Z = 1 + O(\lambda)$, thus at the leading order we can set $Z = 1$ and thus we have

$$\langle f| \hat{S} |i\rangle = \int \mathrm{d}^4 x_1 \,\dots\, \mathrm{d}^4 x_4 e^{i(p_3 x_3 + p_4 x_4 - p_1 x_1 - p_2 x_2)} \times \langle\Omega| \hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\} |\Omega\rangle\,.$$

Now we can use the Feynman rules in position space to write down the contribution of the leading order diagram, which is given by

...diagrams...

$$O(\lambda^0) \;:\; D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23},$$

where we have three possible contractions between the four fields, and thus we have three possible diagrams. Now we can write down the contribution of each diagram in momentum space by using the Fourier transform of the Feynman propagator, which is given assigning a momentum k to the internal line (with definite direction $i \to j$), and thus we have

$$D_{ij} = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{e^{-ik(x_i - x_j)}}{k^2 - m^2 + i\epsilon},$$

[...]

$$\text{Diag I: } \prod_{j=1}^{4} \left(-i(p_j^2 - m^2)\right) \times \delta^{(4)}(p_1 + p_2)\delta^{(4)}(p_3 + p_4) \times \frac{i}{p_1^2 - m^2} \frac{i}{p_3^2 - m^2} = 0,$$

since the external momenta are on-shell, thus $p_j^2 = m^2$ for $j = 1, 2, 3, 4$. For the second diagram we can repeat the same procedure, and we get

$$\text{Diag II: } \prod_{j=1}^{4} \left(-i(p_j^2 - m^2)\right) \times \delta^{(4)}(p_1 - p_3)\delta^{(4)}(p_2 - p_4) \times \frac{i}{p_1^2 - m^2} \frac{i}{p_2^2 - m^2},$$

where the delta functions set $p_3 = p_1$ and $p_4 = p_2$, which is describing the process where the particles p_1 and p_2 do not scatter, leading to a final state that is the same as the initial state (it is the identity process of $\hat{S} = \mathbb{I} + i\hat{T}$). The third diagram gives us the same contribution as the second diagram, since it is just a permutation of the external points, and we can ignore these two processes.

We are interested in the scattering process where the particles p_1 and p_2 interact and scatter into the final state p_3 and p_4 such that $|f\rangle \neq |i\rangle$. The first non trivial diagrams come from the second order correction, which is given by

TODO: draw diagrams

$$O(\lambda^2) \; : \; \ldots [\text{diagrams}] \ldots$$

TODO: draw diag 1 with k-directions

...

$$\text{Diag I: } \prod_{j=1}^{4} \left(-i(p_j^2 - m^2)\right) \times \int \mathrm{d}^4 x_1 \, \ldots \, \mathrm{d}^4 x_n \, e^{i(p_3 x_3 + p_4 x_4 - p_1 x_1 - p_2 x_2)} \times (-i\lambda)^2 \int \mathrm{d}^4 x \, \mathrm{d}^4 y \, D_{1x} D_{2x} D_{xy} D_{y3} D_{y4}$$

$$= \prod (\cdot) \times \int \mathrm{d}^4 x_1 \, \ldots \, \mathrm{d}^4 x_n \times (-i\lambda)^2 \int \frac{\mathrm{d}^4 k_1}{(2\pi)^4} \cdots \frac{\mathrm{d}^4 k_4}{(2\pi)^4} \frac{\mathrm{d}^4 k}{(2\pi)^4} e^{-ix_1(p_1 - k_1)} e^{-ix_2(p_2 - k_2)} e^{-ix_3(p_3 - k_3)} e^{-ix_4(p_4 - k_4)} e^{ix(k - k_1 - k_2}$$

[...] deltas [...]

$$\langle f | \, i\hat{T} \, | i \rangle = i\mathcal{M}(2\pi)^4 \delta^{(4)}(p_{in} - p_{out}),$$

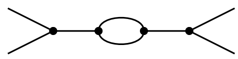
where $\mathcal{M}$ is the scattering amplitude, which is a function of the external momenta, and it encodes all the information about the scattering process, and it is the quantity that we are interested in calculating. To recap:

- comments
- comments
- momentum conservtion

If we conclude the calculation for the first diagram, we get

$$\text{Diag I: } \frac{i}{(p_1 + p_2)^2 - m^2}.$$

The delta functions do not always cancel out all the integrals over momenta, and one can show indeed that if the diagram contains a loop, then we are left with an integral over the loop momentum, which is not fixed by the delta functions, and thus we need to integrate over it. For example, if we consider the following diagram, which is a one-loop correction to the 2-point function:



...

Summary of Rules in Momentum Space

Let us summarize the Feynman rules in momentum space for a scalar field theory:

- $i\mathcal{M} = \sum$ fully connected diagrams, where the sum is over all the fully connected diagrams that contribute to the scattering process we are interested in, and $\mathcal{M}$ is the scattering amplitude, which encodes all the information about the scattering process.
- We assign a momentum $\mathbf{k}$ to each internal line, with the convention that the momentum flows from the initial state to the final state (thus modulus and direction matter).
- The external lines are on-shell, thus they satisfy the mass-shell condition $p_j^2 = m^2$ for $j = 1, 2, 3, 4$ and ...
- The internal lines correspond to propagators, which in momentum space are given by $\frac{i}{k^2 - m^2 + i\epsilon}$, where k is the momentum flowing through the internal line.
- The vertexes correspond to factors from the interaction Lagrangian, which in momentum space are given by $-i\lambda$ for a ϕ^3 theory, and more generally by the Fourier transform of the interaction term in the Lagrangian. They depend on the theory under consideration, and they also depend on the momenta flowing through the vertex.
- **Momentum conservation at each vertex**: the sum of the incoming momenta must equal the sum of the outgoing momenta, which is enforced by delta functions in the mathematical expression corresponding to each diagram.
- Each loop in the diagram corresponds to an integral over the loop momentum, which is not fixed by the delta functions, and thus we need to integrate over it. The measure of the integral is given by $\int \frac{\mathrm{d}^4 k}{(2\pi)^4}$ for each unresolved loop momentum k.
- Symmetry factor: $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram. Same as in position space.

For leading order (LO) corrections, this is plenty enough to understand the scattering process and calculate the scattering amplitude, while for higher order corrections, we need to take into account also the renormalization of the theory, which is a more advanced topic that we will cover in the next chapter.

When saying **fully connected diagrams**, we mean that the diagram is connected and that it *does not contain any disconnected subdiagram.* [...] For example, in the following $2 \to 4$ diagram: [...] The non fully connected diagram describes to separate processes more than a global interaction of 2 incoming particles scattering into 4 outgoing particles, and thus it does not contribute to the scattering amplitude of the process we are interested in, while the fully connected diagram describes exactely the process we are interested in.

Remark. *At leading order we have set $Z = 1$, and we confused the lagrangian mass with the physical one:...*

If we write again the second order correction to the $2 \to 2$ scattering amplitude in momentum space, we get

$$[\ldots diagrams\, and\, momenta \ldots]$$

$$i\mathcal{M} = (-i\lambda)^2 \left[\frac{i}{(p_1+p_2)^2 - m^2} + \frac{i}{(p_1-p_3)^2 - m^2} + \frac{i}{(p_1-p_4)^2 - m^2} \right]$$

where the $(-i\lambda)^2$ factor comes from the two vertexes, common for all the three diagrams, while inside the parenthesis we have the three propagator contributions of each diagram, which is given by the product of the propagators corresponding to the internal lines, and the delta functions that enforce momentum conservation at each vertex have already been integrated out: only applying the rules we got the result which would have implied a lot of calculationsotherwise.

Back to LSZ Reduction Formula

Now consider a $2 \to n$ scattering process, where we have two incoming particles and n outgoing particles. If we want to calculate the scattering amplitude for this process, we can use the LSZ reduction formula, which relates the S-matrix element to the Green's function. The Green's function we need to calculate is given by

$$\ldots$$

drawing of scatter

drawig of diagram with big amputated area

list of: -o- (two point diagram), >o« (amputated diagrams), meaning of amputated

example of amputation

Definition 3.2 (Amputated diagram)**.** An amputated diagram is a Feynman diagram where we cannot disconnect any external line by cutting a single internal line. In other words, an amputated diagram is a diagram that does not contain any disconnected subdiagram, and thus it is fully connected.

The contribution of an amputated diagram to the scattering amplitude is given by the product of the propagators corresponding to the internal lines, and the factors from the vertexes, without including the external propagators, since they are on-shell and thus they do not contribute to the scattering amplitude.

Thus the contribution of the sum of two point diagrams for external lines is given by

$$diog = \sum 2 - point\, diagrams = some\, drowings = \frac{iZ}{P^2 - m^2} + regular\,(p^2 \sim m^2),$$

We are left with a factor of $\sqrt{Z}$ for each external line, which comes from the LSZ reduction formula, and thus we have a total of Z for the two external lines. The amputated diagram is given by

- $i\mathcal{M} = \sum$ fully connected amputated diagrams.
- Assign momentum to internal lines, with the convention that the momentum flows from the initial state to the final state, and it is conserved at each vertex.
- For each unresolved momentum, i.e. for each loop, we need to integrate over it with the measure $\int \frac{\mathrm{d}^4 k}{(2\pi)^4}$.
- External lines contribute with a factor of 1 each, since they are on-shell and thus they do not contribute to the scattering amplitude.
- Internal lines contribute with a factor of $\frac{i}{p^2 - m^2}$ each, where p is the momentum flowing through the internal line: propagator $\equiv$ internal line.
- Vertexes contribute with a factor of $-i\lambda$ for a ϕ^3 theory, and more generally by the Fourier transform of the interaction term in the Lagrangian, which depends on the theory under consideration and on the momenta flowing through the vertex.
- The symmetry factor of the diagram is given by $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram.
- For each external particle, we need to multiply by a factor of $\sqrt{Z}$, which comes from the LSZ reduction formula.

We derived these feynman rules from a real scalar theory with a cubic interaction, but they can be easily generalized to other theories: the only theory specific point is the vertex factor, which depends on the interaction term in the Lagrangian, while the other rules are general and can be applied to any theory.

3.3 | Applications of Feynman Diagrams

Let us now apply the Feynman rules to compute scattering amplitudes for various processes in different quantum field theories. We will start with the simplest case of a scalar field theory and then move on to more complex theories involving fermions and gauge bosons.

We will divide diagrams in two categories:

1. **The tree-level diagrams**: These are the diagrams that do not contain any loops. They represent the leading order contribution to the scattering amplitude and are often the simplest to compute.

TODO: add example drawing

2. **The loop diagrams**: These are the diagrams that contain one or more loops. They represent higher-order corrections to the scattering amplitude and can be more challenging to compute due to the need for regularization and renormalization techniques.

TODO: add example drawing

Thus in the loop diagrams, one or more internal lines form a closed loop, which can lead to unresolved integrals like $\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2-m^2+i\epsilon}$, which is divergent and requires regularization and renormalization to obtain a finite result. (?)

Usually the first type of diagrams (tree-level) gives the dominant contribution to the scattering amplitude, while the second type (loop diagrams) provides corrections that can be important for precision calculations. For the first ones, usually some algebra is enough to compute the amplitude, since we have no integrations left, while for the second ones, one needs to use more advanced techniques to handle the divergences and obtain meaningful results. Loop diagrams can also give rise to interesting phenomena such as virtual particles and vacuum polarization, which can have important implications for the behavior of quantum fields and the structure of spacetime, but they are in general more difficult to solve.

Remark. *We said that usually at LO only the tree diagrams appears, but this is nothing general: there exist cases in which we have loops appearing alredy at the leading order, which are more complicated.*

Example (EXERCISE). Compute the differential cross section $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ for the $2 \to 2$ scatter in ϕ^3 theory at tree level, in the center of mass frame, as a function of the total center of mass energy E and the scattering angle θ.

TODO: add drawing with scattering angle

Example ($2 \to 2$ scattering in ϕ^3 theory). We can define the scattering angle θ as the angle between the incoming particle with momentum p_1 and the outgoing particle with momentum p_3:

$$\cos\theta = \frac{\mathbf{p}_1 \cdot \mathbf{p}_3}{|\mathbf{p}_1||\mathbf{p}_3|}.$$

At the leading order, the matrix element for the $2 \to 2$ scattering process in ϕ^3 theory can be computed using the Feynman rules: the relevant Feynman diagrams at tree level include the s-channel, t-channel, and u-channel diagrams

$$i\mathcal{M} = diagrams + O(\lambda^4).$$

Before proceeding with the calculation, we have to study in more detail the kinematics of the process, which is the subject of the next section.

3.3.1 | 4-Point Kinematics

In a $2 \to 2$ scattering process, we have two incoming particles with momenta p_1 and p_2, and two outgoing particles with momenta p_3 and p_4. We have

$$p_i^2 = m^2,$$

since all particles are on-shell. The scattering amplitude is lorentz invariant, so that it is useful to express it in terms of the **Mandelstam variables** s, t, and u.

Definition 3.3 (Mandelstam variables)**.** The Mandelstam variables s, t, and u, are defined as follows:

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2, \\ t &= (p_1 - p_3)^2 = (p_4 - p_2)^2, \\ u &= (p_1 - p_4)^2 = (p_2 - p_3)^2. \end{aligned}$$

These variables are not independent, since they satisfy the relation

$$s + t + u = 4m^2.$$

They are defined in such a way that they are invariant under Lorentz transformations, and they provide a convenient way to express the scattering amplitude in terms of the kinematic variables of the process.

Now the idea is to express the scattering amplitude in terms of the Mandelstam variables, which will allow us to compute the differential cross section as a function of the scattering angle θ and the total center of mass energy E. Starting from their definitions, we can express

$$\begin{aligned} s = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = m_1^2 + m_2^2 + 2p_1 \cdot p_2&, \\ &\begin{cases} p_1 \cdot p_2 = \dots \\ p_1 \cdot p_3 = \dots \\ p_2 \cdot p_3 = \dots \end{cases} \\ p_4 = p_1 + p_2 - p_3&. \end{aligned}$$

Thus we are going to square the last relation, and we will obtain

$$\begin{aligned} m_4^2 = p_4^2 = (p_1 + p_2 - p_3)^2 &= m_1^2 + m_2^2 + m_3^2 + 2p_1 \cdot p_2 - 2p_1 \cdot p_3 - 2p_2 \cdot p_3 \\ &= m_1^2 + m_2^2 + m_3^2 + 2\tfrac{1}{2}\left(s - m_1^2 - m_2^2\right) - 2\left(-\tfrac{1}{2}\right)\left(t - m_1^2 - m_3^2\right) - 2\left(-\tfrac{1}{2}\right)\left(u - m_2^2 - m_3^2\right) \\ &= m_1^2 + m_2^2 + m_3^2 + s - m_1^2 - m_2^2 + t - m_1^2 - m_3^2 + u - m_2^2 - m_3^2 = s + t + u - m_1^2 - m_2^2 - m_3^2. \end{aligned}$$

Thus we have

$$s + t + u = \sum_j m_j^2,$$

which is the relation we wanted to prove. For an N-particle scattering process, we can define the Mandelstam variables as ($N > 4$):

1. $4N$ momentum components p_i^μ for $i = 1, \dots, N$.

2. N on-shell conditions $p_i^2 = m_i^2$.

3. 4 momentum conservation conditions $\sum_{i=1}^N p_i^\mu = 0$.

4. 6 independent Lorentz invariants $p_i \cdot p_j$ for $i < j$, i.e. 6 lorentz transformations leaving the amplitude invariant.
5. Thus we have $4N - N - 4 - 6 = 3N - 10$ independent Mandelstam variables, which can be chosen as $s_{ij} = (p_i + p_j)^2$ for $i < j$, which are the generalization of the Mandelstam variables for N-particle scattering processes.

Now for a $2 \to 2$ general process, in the center of mass frame, we have two incoming particles parallel along the z-axis, and two outgoing particles with momenta $\mathbf{p}_3$ and $\mathbf{p}_4$ forming an angle θ with the z-axis

$$p_1 = \begin{pmatrix} E_1 \\ 0 \\ 0 \\ p \end{pmatrix}, \quad p_2 = \begin{pmatrix} E_2 \\ 0 \\ 0 \\ -p \end{pmatrix},$$

where $p = |\mathbf{p}_1| = |\mathbf{p}_2|$. Now we consider the outgoing particles, on the z-y-plane, with momenta

$$\mathbf{p}_4 = \mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_3, \quad p' = |\mathbf{p}_3| = |\mathbf{p}_4|,$$

$$\mathbf{p}_3 = \begin{pmatrix} E_3 \\ 0 \\ p' \sin\theta \\ p' \cos\theta \end{pmatrix}, \quad \mathbf{p}_4 = \begin{pmatrix} E_4 \\ 0 \\ -p' \sin\theta \\ -p' \cos\theta \end{pmatrix}.$$

Now let us compute the center of mass energy E and the Mandelstam variables in terms of the scattering angle θ. We have

$$E \equiv E_{cm} = E_1 + E_2 = E_3 + E_4,$$

with the first Mandelstam variable defined as

$$s = (p_1 + p_2)^2 = (E_1 + E_2)^2 = E^2,$$

since the spatial momenta of the incoming particles cancel out. Now

$$(p_1 + p_2) \cdot p_1 = E\, E_1,$$

since

$$(p_1 + p_2) = m_1^2 + p_1 \cdot p_2 = m_1^2 + \frac{1}{2}(s - m_1^2 - m_2^2) = \frac{1}{2}(E^2 + m_1^2 - m_2^2).$$

Thus we have

$$\begin{cases} E_1 = \dfrac{E^2 + m_1^2 - m_2^2}{2E}, \\ E_2 = \dfrac{E^2 + m_2^2 - m_1^2}{2E}, \\ E_3 = \dfrac{E^2 + m_3^2 - m_4^2}{2E}, \\ E_4 = \dfrac{E^2 + m_4^2 - m_3^2}{2E}, \end{cases}$$

where the last two relations can be obtained by doing the same calculation for the outgoing particles.

Now

$$m_1^2 = p_1^2 = E_1 - |\mathbf{p}_1|^2, \quad p = \frac{\sqrt{[E^2 - (m_1 + m_2)^2]\,[E^2 - (m_1 - m_2)^2]}}{2E},$$

and simiarly for the outgoing particles p'. For the second Mandelstam variable, we have

$$t = (p_1 - p_3)^2 = m_1^2 + m_3^2 - 2p_1 \cdot p_3 = m_1^2 + m_2^2 - 2(E_1 E_2 - pp' \cos\theta) = t(E, \theta).$$

Special cases

$(m_1 = m_2)$. In this case, we have

$$E_1 = E_2 = \frac{E}{2}, \quad p = \frac{\sqrt{E^2 - 4m^2}}{2}.$$

$(m_3 = m_4)$. In this case, we have

$$E_3 = E_4 = \frac{E}{2}, \quad p' = \frac{\sqrt{E^2 - 4m^2}}{2}.$$

Elstic scattering. In this case, we have $m_1 = m_3$ and $m_2 = m_4$, so that

$$E_1 = E_3, \quad E_2 = E_4, \quad p = p'.$$

$(m_1 = m_2 = m_3 = m_4)$. In this case, we have

$$E_1 = E_2 = E_3 = E_4 = \frac{E}{2}, \quad p = p' = \frac{\sqrt{E^2 - 4m^2}}{2}.$$

This is the combination of the previous two cases, and it is the most common one in practice, since it describes the scattering of identical particles. The mandelstam variables in this case are given by

$$\begin{cases} s = E^2, \\ t = -\frac{1}{2}(E^2 - 4m^2)(1 - \cos\theta), \\ u = -\frac{1}{2}(E^2 - 4m^2)(1 + \cos\theta), \end{cases}$$

with the usual relation $s + t + u = 4m^2$.

Now we have what we need to compute the scattering amplitude for the $2 \to 2$ scattering process in ϕ^3 theory at tree level, which is given by the sum of the s-channel, t-channel, and u-channel diagrams. Let us come back to this example.

Example ($2 \to 2$ scattering in ϕ^3 theory). Remember that at the leading order, the matrix element for the $2 \to 2$ scattering process in ϕ^3 theory can be computed using the Feynman rules: the relevant Feynman diagrams at tree level include the s-channel, t-channel, and u-channel diagrams, which have the following momentum dependence:

$$i\mathcal{M} = \mathit{diagrams}(p_1, p_2, p_3, p_4) + O(\lambda^4)$$

which contribute to the scattering amplitude as follows:

$$i\mathcal{M} = (-i\lambda)^2 \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right] + O(\lambda^4),$$

from which it is easy to understand why they are called s-channel, t-channel, and u-channel diagrams. Now we can express the scattering amplitude in terms of the Mandelstam variables (expressing them in terms of E and θ as we found before), which will allow us to compute the differential cross section as a function of the scattering angle θ and the total center of mass energy E.

In particular, we have the differential cross section found at the beginning of the section, which in the center of mass frame can be expressed as

$$\begin{aligned}\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \frac{1}{64\pi^2 s}\frac{p'}{p}|\mathcal{M}|^2 = \frac{1}{64\pi^2 E^2}\frac{\sqrt{E^2-4m^2}}{\sqrt{E^2-4m^2}}|\mathcal{M}|^2 \\ &= \frac{1}{64\pi^2 E^2}|\mathcal{M}|^2 = \frac{\lambda^4}{64\pi^2 E^2}\left|\frac{1}{s-m^2}+\frac{1}{t-m^2}+\frac{1}{u-m^2}\right|^2 + O(\lambda^6).\end{aligned}$$

3.3.2 | Feynman Rules for Other Theories

3.4 | General Approach for Feynman Rules

3.4.1 | Complex Scalar Field

3.4.2 | Dirac Fermion Field

Additional Feynman Rules for Fermions

Example (Yukawa theory).

Example.

3.4.3 | Vector Bosons

3.4.4 | Quantum Electrodynamics (QED)

Example (Two different types of particles in QED).

3.4.5 | Observations on Diagrams

3.5 | Crossing Symmetry

3.5.1 | Crossing Symmetry and Identical Fermions

4 | Quantum Electrodynamics - QED

4.1 | Scattering Processes in QED

4.1.1 | Electron - Muon Scattering

4.1.2 | Dirac Algebra

Example.

4.1.3 | Back to Electron - Muon Scattering

4.1.4 | Polarized Matrix Elements

4.1.5 | Compton Scattering and External Photons

Polarization Sums

Compton Scattering

4.1.6 | Notes on Convergence

4.1.7 | Ward Identity and Lorentz Transformations

4.2 | QED as an Abelian Gauge Theory

4.3 | Born Approximation and Non-Relativistic Scattering

Appendices

A | Notation and Conventions

Conventions on Units and Natural Units

We are going to use natural units throughout these notes, where the speed of light c and the reduced Planck constant $\hbar$ are set to 1:

$$c = 1, \quad \hbar = 1.$$

In natural units, all physical quantities are expressed in terms of a single unit of energy (or mass) and measured in Electronvolts (eV) and multiples thereof, since the dimensions of length and time are related to energy through the relations $E = mc^2$ and $E = \hbar\omega$. This simplifies many equations in quantum field theory and allows us to focus on the essential physics without being distracted by factors of c and $\hbar$. In these units, we have:

$$\begin{aligned}
[E] &= [M] = [L]^{-1} = [T]^{-1}, \\
[\hbar] &= 1, \\
[c] &= 1.
\end{aligned}$$

We report some useful conversion factors between natural units and SI units:

$$\begin{aligned}
1\,\mathrm{GeV} &\approx 1.602 \times 10^{-10}\,\mathrm{J}, \\
1\,\mathrm{GeV} &\approx 1.783 \times 10^{-27}\,\mathrm{kg}, \\
1\,\mathrm{GeV}^{-1} &\approx 0.1975\,\mathrm{fm}, \\
1\,\mathrm{GeV}^{-1} &\approx 6.582 \times 10^{-25}\,\mathrm{s},
\end{aligned}$$

and also

$$\begin{aligned}
\hbar c &\approx 197.3\,\mathrm{MeV} \cdot \mathrm{fm}, \\
1\,\mathrm{fm} &\approx 10^{-15}\,\mathrm{m},
\end{aligned}$$

where the femtometer (fm) is a common unit of length in nuclear and particle physics, often called a Fermi: it is approximately the length of a proton or neutron radius:

$$r_p \sim 0.8\,\mathrm{fm} = \frac{0.8}{197.3}\,\mathrm{MeV}^{-1}.$$

Conventions on Electromagnetism

In order to avoid confusion, we will use the **Heaviside-Lorentz** units for electromagnetism throughout these notes: in these units, the electric charge e is dimensionless and appears explicitly in the interaction terms of the Lagrangian, making it easier to identify the strength of electromagnetic interaction, while Maxwell equations take a more symmetric form.

Explicitly, in Heaviside-Lorentz units, we have the following relations:

- $\epsilon_0 = \mu_0 = 1$, where ϵ_0 is the *vacuum permittivity* and μ_0 is the *vacuum permeability*: this means that the electric and magnetic fields have the same units and that the speed of light c is equal to 1 (these units are consistent with the natural units we are using).
- The **electric charge** e is dimensionless and connected to the fine-structure constant α by
$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2}{4\pi} \approx \frac{1}{137}.$$
- The **electric potential** Φ is defined as
$$\Phi = \frac{Q}{4\pi R},$$
where Q is the charge and R is the distance from the charge: this means that the electric potential has the same units as the electric field E (which is consistent with the fact that in Heaviside-Lorentz units, $\epsilon_0 = 1$).
- The **Maxwell equations** take the following form:
$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t}, \end{aligned}$$
where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the charge density, and $\mathbf{J}$ is the current density: this symmetric form of Maxwell equations is one of the reasons why Heaviside-Lorentz units are often preferred in theoretical physics, especially in quantum field theory. Basically the Heaviside-Lorentz units are a rationalized version of the Gaussian units, where the factors of 4π are absorbed into the definitions of the fields and charges, leading to simpler equations and making it easier to identify the physical constants that govern the interactions.

Conventions on the Metric and Spacetime

We will use the **mostly minus** convention for the metric tensor $g_{\mu\nu}$ in Minkowski spacetime, which is defined as
$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$
This means that the time component of the metric is positive, while the spatial components are negative; using Einstein summation convention, we have the following relations among the components of the metric and v^μ, w^μ four vectors:
$$\begin{aligned} v \cdot w &= g_{\mu\nu} v^\mu w^\nu = v^0 w^0 - \mathbf{v} \cdot \mathbf{w}, \\ v^2 &= v \cdot v = g_{\mu\nu} v^\mu v^\nu = (v^0)^2 - \mathbf{v}^2, \\ v_\mu = g_{\mu\nu} v^\nu &\implies v_0 = v^0, \quad \mathbf{v} = v^i = -v_i. \end{aligned}$$

We will use greek letters μ, ν, ρ, σ to denote spacetime indices that run from 0 to 3, while Latin letters i, j, k will be used to denote spatial indices that run from 1 to 3:

$$\mu, \nu, \rho, \sigma \in \{0,\ 1,\ 2,\ 3\} = \{t,\ x,\ y,\ z\},$$
$$i, j, k \in \{1,\ 2,\ 3\} = \{x,\ y,\ z\}.$$

Fourier Transforms

We will use the following conventions for Fourier transforms of a function $f(x)$ in four-dimensional spacetime:

$$f(x) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} e^{-ip\cdot x} \tilde{f}(p),$$
$$\tilde{f}(p) = \int \mathrm{d}^4 x e^{ip\cdot x} f(x).$$

We highlight that the Fourier transform of a function $f(x)$ is denoted by $\tilde{f}(p)$, and that the integration measure includes a factor of $(2\pi)^{-4}$ in the inverse Fourier transform, which is a common convention in quantum field theory: in the space of coordinates, we have the measure d^4x, while in the momentum space, we have the measure $\frac{\mathrm{d}^4 p}{(2\pi)^4}$. This convention ensures that the Fourier transform and its inverse are consistent and that the normalization of the fields and operators in quantum field theory is correct.

Feynman Diagrams

In these notes, we will use Feynman diagrams as a graphical representation of the interactions between particles and fields in quantum field theory. We will use the following conventions for Feynman diagrams:

TODO: Check and change in the end

- Time will be represented on the horizontal axis, going from left to right, which is a common convention in quantum field theory; this means that the left side of the diagram will represent the initial state of the particles, while the right side will represent the final state after the interaction has occurred.

- Solid lines will represent fermions (spin-1/2 particles), while dashed lines will represent bosons (spin-0 or spin-1 particles).

- Arrows on the lines will indicate the direction of particle flow, with arrows pointing towards the interaction vertex for incoming particles and away from the vertex for outgoing particles.

- Wavy lines will represent gauge bosons (spin-1 particles that mediate interactions), such as photons, W and Z bosons, and gluons.

- The vertices in the diagrams will represent interaction points where particles interact according to the rules of the theory, with each vertex corresponding to a specific term in the Lagrangian of the quantum field theory.

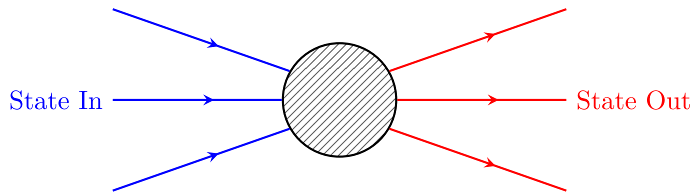


Figure 1: Generic Feynman diagram representing an interaction between particles, where the blue lines represent incoming particles (initial state) and the red lines represent outgoing particles (final state). The black box represents the interaction vertex, which corresponds to a specific term in the Lagrangian of the quantum field theory.

B | Computations and further developments

Differential Cross Section

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